SUPPLEMENT TO "MECHANISTIC MODELS FOR PANEL DATA: ANALYSIS OF ECOLOGICAL EXPERIMENTS WITH FOUR INTERACTING SPECIES"

BY BO YANG[1,a], JESSE WHEELER[2,b], MEGHAN A. DUFFY[3,c], AARON A. KING[3,d] AND EDWARD IONIDES[4,e]

[1]*Department of Marketing, Columbia University,* [a]*ybb@umich.edu*

[2]*Department of Mathematics and Statistics, Utah State University,* [b]*jesse.wheeler@usu.edu*

[3]*Department of Ecology and Evolutionary Biology, University of Michigan,* [c]*duffymeg@umich.edu;* [d]*kingaa@umich.edu*

[4]*Department of Statistics, University of Michigan,* [e]*ionides@umich.edu*

CONTENTS

S1. Monte Carlo Adjusted Profile (MCAP). Monte Carlo Adjusted Profile (MCAP) methods (Ionides et al., 2017; Ning et al., 2021) provide confidence intervals for situations where the likelihood is evaluated and maximized by Monte Carlo algorithms. A smoothed estimate of the profile likelihood is used to reduce Monte Carlo error, quantify this error, and adjust the confidence intervals accordingly to maintain their coverage.

MCAP is particularly useful for high-dimensional situations, such as arise in panel data analysis, since it becomes practically impossible to apply sufficient computational effort to make Monte Carlo error negligible. MCAP has previously been demonstrated for panel iterated filtering by Bretó et al. (2020).

To investigate identifiability, we carried out an additional exploration to obtain estimates, profiles and confidence intervals for the composite parameters, $r^{\Bbbk} f_S^{\Bbbk}$ and $p^{\Bbbk} f_S^{\Bbbk}$, as well as the individual parameters. We could have reparameterized to apply MCAP to these composite parameters exactly as we did for the individual parameters. However, for this additional analysis we made a simplification by re-using the calculations for the original profiles. For every point on every profile, computed for the individual parameters, we calculated the value of the composite parameters. We obtained the highest log-likelihoods within a grid of values for the composite parameters and applied MCAP to these values. We have found previously that these "poor man's profiles" provide a reasonable approximation to the true profile, while avoiding the need to work with a reparameterized version of the model. We calculated these composite parameters only for the SIRJPF2 model, and the results are shown together with the individual parameters, in Table 1 and Figure S-1.

S2. The Panel Iterated Filtering Algorithm. Algorithm S1 presents pseudocode for PIF. Here, $\Phi^{F,m}_{u,n,j}$ is the jth filter particle searching for the MLE of the shared parameter ϕ in the mth filter iteration, at time point n and unit u. $\Psi^{F,m}_{u,n,j}$ is the corresponding quantity searching for the MLE of the unit-specific parameter, ψ_u. Cooling factor ρ determines how aggressively the parameter perturbations are dampened as the number of iterations increases. By gradually reducing the size of these parameter random walks, the algorithm refines the parameter estimates and converges more tightly to the maximum likelihood point.

When $\mathrm{MARGINALIZE} = \mathrm{TRUE}$, the estimate of the unit-specific parameter ψ_u belonging to particle j is unchanged when filtering through a unit $\tilde{u} \neq u$. The unmarginalized PIF was proposed by Bretó et al. (2020) and was provided with theoretical convergence results that are not yet available for MPIF. However, when many parameters are unit-specific, MPIF converges to the maximum likelihood faster than the standard filter, because it does not resample a unit's parameters using data from other units. Theoretical support for MPIF will be published elsewhere (Wheeler et al., 2025).

To briefly review, the model is specified by the initial condition, $f_{X_{u,0}}(\,\cdot\,;\phi,\psi_u)$, the latent state transition density, $f_{X_{u,n}|X_{u,n-1}}\big(\,\cdot \mid ;\phi,\psi_u\big)$, and the measurement density, $f_{Y_{u,n}|X_{u,n}}\big(\,\cdot \mid \cdot\,;\phi,\psi_u\big)$, for $u = 1\!:\!U$ and $n = 1\!:\!N_u$. The full parameter vector is written as $\theta = (\phi, \psi_{1:U})$. For notational convenience, we define $u = 0$ to be an empty panel with $N_0 = 0$. Algorithm S1 supposes Gaussian perturbations of the parameters, which is a common practical choice but may require re-parameterizing to avoid boundaries. For example, non-negative parameters are typically log-transformed. Additional generality is provided by the pseudocode in Bretó et al. (2025) that is implemented in the `mif2` function of the R package `panelPomp`.

Algorithm S1: Panel iterated filter (PIF)

Input: PanelPOMP model, as defined in Section 3 of the main text
Number of particles, J, and filter iterations, M
Cooling factor ρ
Starting parameter vector swarm, $\Theta_j^{(0)} = (\Phi_j^{(0)}, \Psi_{1:U,j}^{(0)})$, for $j = 1\!:\!J$
Perturbation variance, $V_{u,n}^{\Theta}$, for $u = 1\!:\!U$, $n = 1\!:\!N_u$
Logical variable determining marginalization, MARGINALIZE
Output: Parameter swarm, $\Theta_{1:J}^{(M)}$, approximating the MLE

for $m = 1$ **to** M **do**
  $\Theta_{0,j}^{F,m} = \Theta_j^{(m-1)}$ for $j = 1\!:\!J$
  for $u = 1$ **to** U **do**
    $(\Phi_{u,0,j}^{F,m}, \Psi_{u,0,j}^{F,m}) \sim \mathcal{N}(\Theta_{u-1,j}^{F,m}, \rho^{2m} V_{u,0}^{\Theta})$ for $j = 1\!:\!J$
    $X_{u,0,j}^{F,m} \sim f_{X_{u,0}}(\,\cdot\,; \Phi_{u,0,j}^{F,m}, \Psi_{u,0,j}^{F,m})$ for $j = 1\!:\!J$
    for $n = 1$ **to** N_u **do**
      $\Phi_{u,n,j}^{P,m} \sim \mathcal{N}\big(\Phi_{u,n-1,j}^{F,m}, \rho^{2m} V_{u,n}^{\Phi}\big)$ for $j = 1\!:\!J$
      $\Psi_{u,n,j}^{P,m} \sim \mathcal{N}\big(\Psi_{u,n-1,j}^{F,m}, \rho^{2m} V_{u,n}^{\Psi}\big)$ for $j = 1\!:\!J$
      $X_{u,n,j}^{P,m} \sim f_{X_{u,n}|X_{u,n-1}}\big(\,\cdot \mid X_{u,n-1,j}^{F,m}\,; \Phi_{u,n,j}^{P,m}, \Psi_{u,n,j}^{P.m}\big)$ for $j = 1\!:\!J$
      $w_{u,n,j}^{m} = f_{Y_{u,n}|X_{u,n}}\big(y_{u,n}^{*} \mid X_{u,n,j}^{F,m}\,; \Phi_{u,n,j}^{P,m}, \Psi_{u,n,j}^{P.m}\big)$ for $j = 1\!:\!J$
      $k_j = i$ with probability proportional to $w_{u,n,i}^{m}$ for $i, j = 1\!:\!J$
      $X_{u,n,j}^{F,m} = X_{u,n,k_j}^{P,m}$ for $j = 1\!:\!J$
      $\Phi_{u,n,j}^{F,m} = \Phi_{u,n,k_j}^{P,m}$ for $j = 1\!:\!J$
      $\Psi_{u,n,j}^{F,m} = \Psi_{u,n,k_j}^{P,m}$ for $j = 1\!:\!J$
      if MARGINALIZE **then**
        $\Psi_{\tilde{u},n,j}^{F,m} = \Psi_{\tilde{u},n,j}^{P,m}$ for all $\tilde{u} \neq u$, $j = 1\!:\!J$
      else
        $\Psi_{\tilde{u},n,j}^{F,m} = \Psi_{\tilde{u},n,k_j}^{P,m}$ for all $\tilde{u} \neq u$, $j = 1\!:\!J$
      end
    end
    $\Theta_{u,j}^{F,m} = \big(\Phi_{u,N_u,j}^{F,m}, \Psi_{1:U,N_u,j}^{F,m}\big)$
  end
  $\Theta_j^{(m)} = \Theta_{U,j}^{F,m}$;
end

S3. MCAP Coverage Study. To see how well the MCAP confidence intervals perform in finite samples, we carried out a parametric bootstrap. We simulated $B = 100$ datasets from the fitted all-shared SIRJPF2 maximum likelihood estimate, and on each one we recomputed the MCAP profile and asked whether its 95% interval contained the value that generated the data. We did this for four parameters chosen to span the range of identifiability: the native and invasive birth rates $r^{\mathbb{n}}$ and $r^{\mathbb{i}}$, the parasite-spore process noise σ_P, and the food process noise σ_F.

A profile confidence interval collects the parameter values whose profile log-likelihood lies within a cutoff c of the maximum, $\{\theta : \ell^{\max} - \ell(\theta) \leq c\}$. The basic MCAP interval uses the asymptotic cutoff $c = \chi^2_{1,0.95}/2 = 1.92$. Where this under-covers, we also report a corrected interval that replaces 1.92 by the cutoff needed to reach 95% coverage in the simulation, namely the 95th percentile of the profile drop at the truth, $D = \ell^{\max} - \ell(\theta_{\text{true}})$. A larger cutoff widens the interval, so the corrected interval is simply the same profile cut at a higher threshold. Table S1 reports both.

The invasive birth rate $r^{\mathbb{i}}$ is a special case. It enters the process model only through the product $r^{\mathbb{i}} f^{\mathbb{i}}_S$, just as $p^{\mathbb{i}}$ enters only through $p^{\mathbb{i}} f^{\mathbb{i}}_S$, so it is weakly identifiable, with a profile that is essentially flat. It has a nominal 94.0% coverage, since some extreme simulations have confidence intervals that fail to cover the truth, but the parameter is better classified as practically non-identifiable.

TABLE S1

Parametric-bootstrap coverage of the MCAP 95% confidence intervals, on the log scale, across $B = 100$ datasets simulated from the all-shared SIRJPF2 maximum likelihood estimate. The basic interval uses the asymptotic cutoff $c = 1.92$ and matches Table 1 of the main text; the corrected interval re-cuts the same profile at the calibrated cutoff c that targets 95% coverage. The native birth rate $r^{\mathbb{n}}$ is biased a little high and the noise parameters σ_P and σ_F a little low, all on weakly identified ridges. The invasive birth rate $r^{\mathbb{i}}$ has a flat profile and is not separately identifiable, so we omit it from the correction. Coverage rates are binomial proportions over the $B = 100$ simulated datasets, so they carry a Monte Carlo standard error of at most about 0.05.

	Truth	MCAP	Basic ($c = 1.92$)		Corrected		
Parameter	(log)	MLE (log)	95% CI	cov.	cutoff c	95% CI	cov.
$r^{\mathbb{n}}$	3.71	4.03	[3.27, 5.27]	88.0%	2.79	[3.12, 5.71]	95.0%
σ_P	−1.30	−1.59	[−3.12, −0.87]	87.0%	3.91	[−3.52, −0.56]	95.0%
σ_F	−1.94	−2.17	[−2.92, −1.46]	76.0%	4.77	[−4.11, −1.06]	95.0%
$r^{\mathbb{i}}$	12.28	flat	non-identifiable; nominal 94.0% coverage degenerate				

The corrected MCAP cutoff has the target 95% coverage by construction, on the simulations used to build this cutoff. To check that the correction captures something real and is not over-fitted to the simulations, we recalibrated the cutoff on a random half of the datasets and measured coverage on the held-out half; averaged over 500 such splits this gives about 93% out-of-sample coverage. We therefore read the coverage study as a reliable exploratory description of uncertainty rather than a guarantee of 95% coverage for downstream decisions. In practical terms, Table S1 shows that reaching nominal coverage in this richly parameterized model takes intervals roughly 1.3 to 2.1 times wider than the asymptotic ones. It was not practical to compute the corrected confidence intervals for every parameter in all the models. We therefore report the basic MCAP interval in Table 1, with the understanding that more precise results are available for parameters of particular interest.

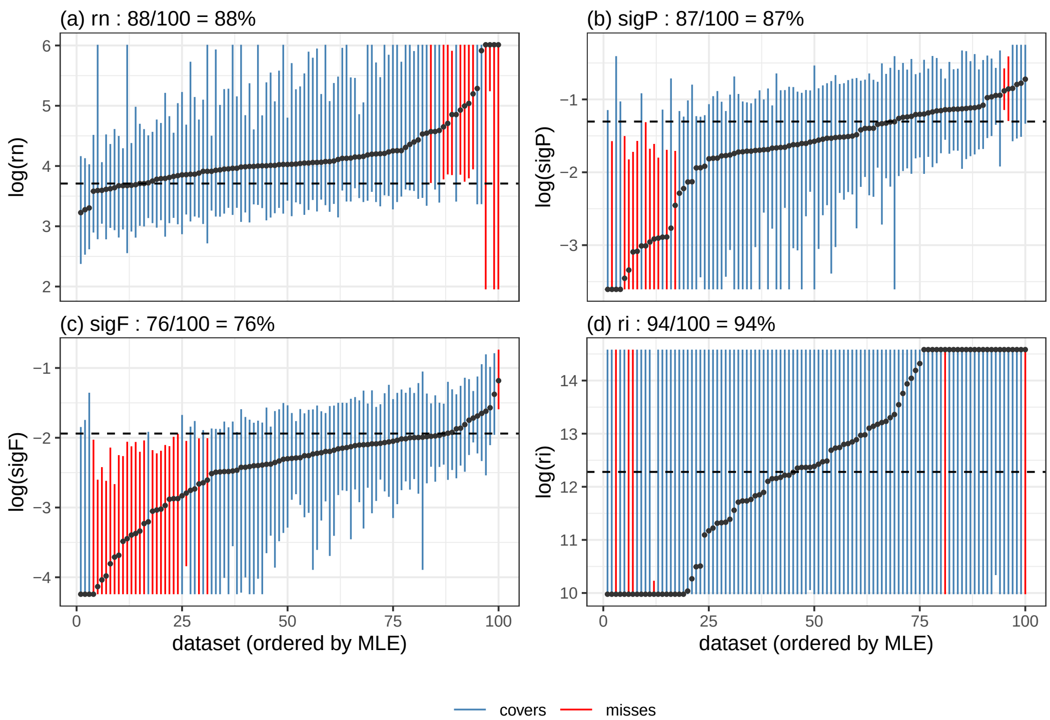


Figure S1: Empirical coverage of the MCAP 95% confidence intervals across $B = 100$ bootstrap datasets simulated from the all-shared SIRJPF2 maximum likelihood estimate, for the four target parameters. In each panel the horizontal lines are the simulated datasets' MCAP intervals, ordered by midpoint, and the vertical line marks the true value; an interval covers if it crosses that line. Panels (a)–(c) are the identifiable parameters, whose undercoverage comes from the maximum likelihood estimate being displaced along weakly identified ridges: r^{n} a little high, σ_P and σ_F a little low, with the largest displacement and the lowest coverage for σ_F. Panel (d) is the r^{i} profile, which is flat because r^{i} enters the process model only through the recruitment product $r^{\mathrm{i}} f_S^{\mathrm{i}}$.

S4. Model selection: Simulation study of AIC and LRT. We conducted simulation studies to check the finite-sample behavior of Akaike's Information Criterion (AIC) and the likelihood ratio test (LRT) for model selection, in our context.

We use LRTs for model comparisons of unit-specific parameters against the nested alternative of shared parameters, in Tables S3, S6, S7, S9, S12, S13. For each of these tests, we compare a p-value from a chi-square approximation against a simulation-based parametric bootstrap. In some situations, we find substantial discrepancy between these two values, and in those cases we give more credence to the finite-sample simulation-based approach. Throughout these tables, the maximized log-likelihoods are Monte Carlo estimates with standard error of order 0.35 (main text), so differences in AIC of only a few units are within Monte Carlo error and are not treated as decisive. Section S5 explains why this may arise, via a simple toy example.

We use AIC to compare log-likelihoods of non-nested models, in Table S3. AIC also provides a quick reference point for comparing nested models. AIC is not designed to be a formal hypothesis test; its goal is to select a good predictive model. Nevertheless, whenever one selects a model, it can be helpful to calculate the sensitivity and specificity of the approach. We check this for the native infected-host mortality rate $\theta_I^{\mathfrak{m}}$, the one parameter for which unit-specific variation was favored by AIC and a chi-square LRT approximation. We simulate panels under two truths: one in which every parameter is shared, at the fitted all-shared SIRJPF2 MLE estimate, and one in which $\theta_I^{\mathfrak{m}}$ is unit-specific, set to the MLE under the model where all other parameters are shared between units. For each simulated panel we fit both the all-shared model and the $\theta_I^{\mathfrak{m}}$-unit-specific model with the same iterated-filtering protocol used in the main analysis, compute the AIC of each, and record which model AIC prefers. Table S2 presents proportions of model selections via AIC in 100 simulations, so the diagonal terms give the rate at which AIC recovers the truth under these hypotheses. Each data-generating truth is simulated 50 times, so the tabulated proportions carry a binomial standard error of at most about 0.07.

Data-generating truth	AIC selects all-shared	AIC selects $\theta_I^{\mathfrak{m}}$ unit-specific
All-shared	0.78	0.22
$\theta_I^{\mathfrak{m}}$ unit-specific	0.08	0.92

TABLE S2

Specificity of AIC when used to choose between shared and unit-specific models

Under the all-shared truth AIC recovered the correct all-shared model on 78% of panels (mean $\Delta\text{AIC} = +5.0$ in its favour), over-selecting the $\theta_I^{\mathfrak{m}}$-unit-specific elaboration on the remaining 22%; under the unit-specific truth it recovered the unit-specific model on 92% of panels (mean $\Delta\text{AIC} = -12.9$). Thus, despite the penalty that AIC places on the more complex model, the method demonstrates some preference toward it. For a formal model selection inference, a simulation-based bootstrap likelihood ratio test is therefore preferable, when it is available.

S5. Discussion about the failure of chi-square LRT. We were initially surprised when the simulation study showed that the likelihood ratio test (LRT) statistic had considerable deviation from the chi-square distribution suggested by a Wilks approximation. To validate this finding, and to investigate whether it might be a general property of a class of POMP models, we developed a simple toy model demonstrating this behavior. Consider a model

$$Y_{un} = \mu_u + \eta_u + \epsilon_{un},$$

for $u = 1, \ldots, U$ and $n = 1, \ldots, N$, with $\epsilon_{un} \sim \mathrm{N}[0, \sigma^2]$ and $\eta_u \sim \mathrm{N}[0, \tau^2]$. This is a mixed effect model with U units, each with N measurements, where each unit mean may have a random effect and/or a fixed effect. The null hypothesis of a shared mean, $H_0 : \mu_u = \mu$, is to be tested against the unit-specific alternative H_1 for which μ_u is unconstrained. The random effect η_u is a simplification of the process noise in a POMP model, and ϵ_{un} is a simple measurement error model. Under H_0 and H_1, the likelihood of data y^*_{un} can be obtained directly by a multivariate normal calculation. Under H_1, with the unit means μ_u free, the random-effect variance is estimated at its boundary, $\hat{\tau} = 0$, for every U and N, because the free fixed effects absorb the between-unit variation that the random effect would otherwise explain. The alternative therefore does not recover the data-generating τ, and it is this boundary behaviour, not the shared structure of the null, that voids the regularity conditions behind the chi-square limit.

General theory for the likelihood ratio test gives a chi-square distribution under local asymptotic normality (Van der Vaart, 2000, Theorem 16.7), without necessarily requiring independence (Geyer, 1994). However, these results require a consistent MLE under H_1. Chi-square approximations to likelihood ratio test statistics have been found to be useful even in some situations where the theoretical asymptotic limit does not apply (Anisimova et al., 2001). In our situation, when τ is large, we may expect that the log-likelihood advantage of using a fixed effect becomes large and so the chi-square approximation becomes poor. Empirically, this is the case, and Figure S2 shows that even a moderate value of τ can give rise to a substantial discrepancy between the actual null distribution and the chi-square approximation.

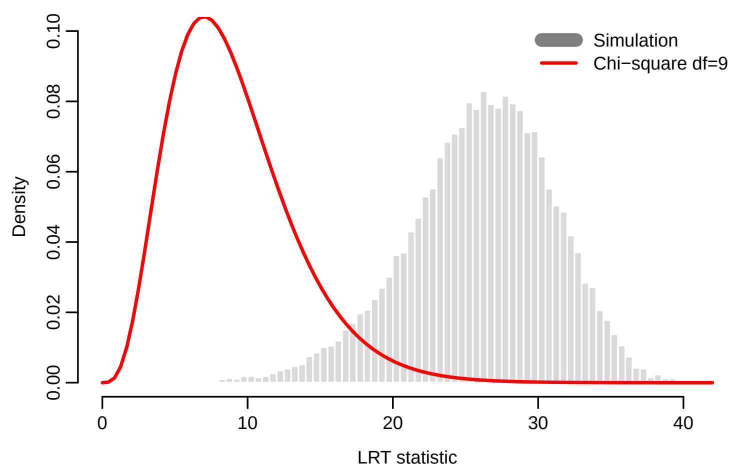


Figure S2: Histogram of the likelihood ratio statistic (twice the difference of the maximized log-likelihoods) for simulations from H_0 with $U = 10$, $N = 20$, $\sigma = 1$, $\tau = 0.5$. The red line shows the chi-square approximation based on Wilks' Theorem.

We conclude that chi-square approximations to the null distribution of the likelihood ratio test statistic are anti-conservative in this context, biasing inference against the null hypothesis. For likelihood ratio tests of shared versus unit-specific parameters, in the presence of random effects and therefore in the presence of dynamic process noise in POMP models, this example suggests the use of a simulation-based null distribution in preference to a chi-square approximation.

S6. SIRJPF2: The Full Model for Two *Daphnia* Species.

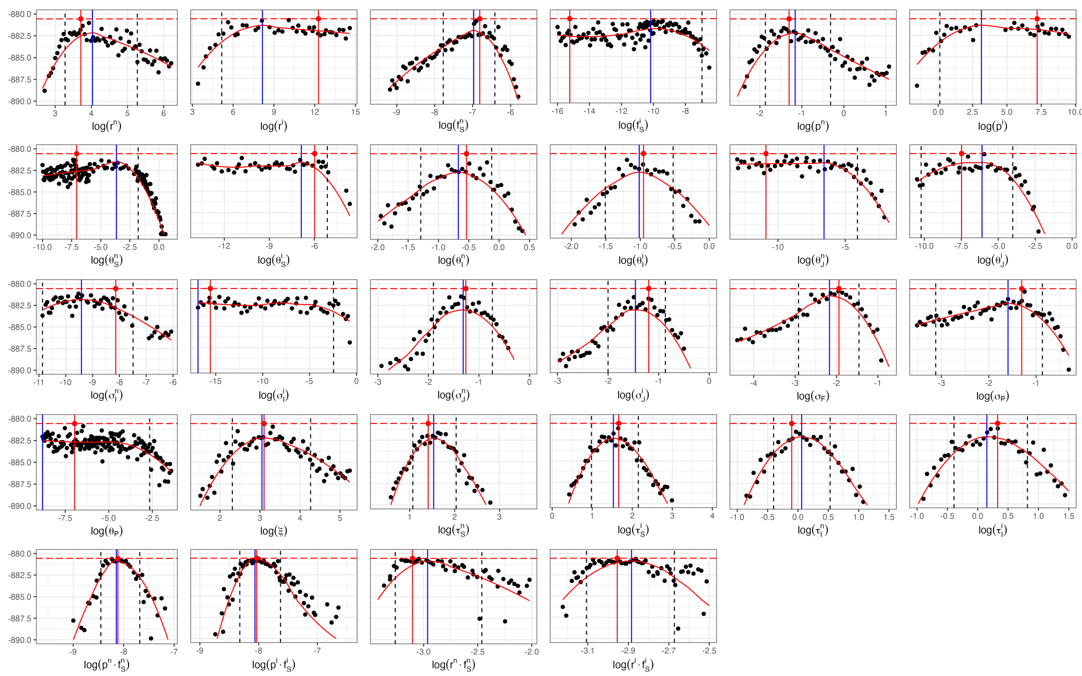


Figure S3: Monte Carlo Adjusted Profile results of SIRJPF2 model for mixed species Parasitized dynamics. The vertical dotted lines represent the 95% confidence interval obtained by MCAP. The vertical blue lines show the MLE estimated using MCAP. The red vertical lines correspond to the model with the overall highest likelihood among all searches.

S6.1. *Results.* In the panelPOMP framework, one must decide whether or not each parameter should be unit specific, or shared across units. These decisions greatly affect the degree of freedom for each model. For example, with ten units in the model, adding a shared parameter increases the model's dimension by one, and adding a unit specific parameter will increase the dimension by ten. An increase in parameter dimension increases the model's ability to quantitatively describe the observed data, but leads to the possibility of over-fitting, resulting in a model with poor explanatory and predictive power. In order to find a balance between fitting accuracy and simplicity when selecting models, we employed the Akaike information criterion (AIC) as a criterion to compare models while adjusting for dimension. We estimate model parameters via maximum likelihood using the panel iterated filtering (PIF) (Bretó et al., 2020) and calculate AIC for all models. In addition to AIC, we report p-values from the likelihood ratio test (LRT) to provide a complementary assessment of whether unit-specific parameterizations yield a statistically significant improvement in model fit over the all-shared baseline. The LRT compares each unit-specific model (alternative) against the all-shared model (null) as a nested pair: the all-shared model is the restricted case obtained by constraining all unit-specific copies of a parameter to be equal. The test statistic $\Lambda = 2[\ell(\text{alternative}) - \ell(\text{null})]$ is conventionally referred to a χ^2 distribution (Wilks' theorem) with degrees of freedom equal to the difference in the number of estimated parameters between the two models, and we report this χ^2 p-value in the tables. However, the regularity conditions for Wilks' theorem do not necessarily hold for our panelPOMP models: the latent-state dynamics are non-standard, some parameters may lie near boundaries, and the log-likelihoods themselves are evaluated by particle filtering and so are subject to Monte Carlo noise. To address this, we additionally report a *simulation-based* (parametric bootstrap) LRT p-value that does not rely on the χ^2 reference distribution. For each comparison we simulate $B = 100$ datasets from the fitted all-shared (null) model, refit both the null and the unit-specific alternative model to each simulated dataset, form the bootstrap test statistic $\Lambda^{(b)} = 2[\ell_{\text{alt}}(y^{(b)}) - \ell_{\text{null}}(y^{(b)})]$, and compute $p_{\text{boot}} = (1 + \#\{b : \Lambda^{(b)} \geq \Lambda_{\text{obs}}\})/(1 + B)$. Table S3 shows the comparison of different models under different settings of unit specific parameter. Each row represents a different model configuration, with specific parameters designated as either unit-specific or shared across all units. Within the table, the max log-likelihood and AIC columns report the maximum log-likelihood and AIC values for each model configuration, where a lower AIC indicates a more favorable balance between model fit and complexity. Similarly, the max log-likelihood (MPIF) and AIC (MPIF) columns provide analogous metrics under the marginalized method setting for the PIF procedure. The LRT p-value column reports the p-value from the likelihood ratio test of each unit-specific model against the all-shared baseline, with "—" denoting the reference model.

Specific parameters	Max loglik (PIF)	AIC (PIF)	Max loglik (MPIF)	AIC (MPIF)	LRT p-value (χ^2)	LRT p-value (bootstrap)
$\varnothing$	-880.56	1809.12	-880.56	1809.12	—	—
$\theta^{\mathbb{k}}_{I,u}$	-867.76	1811.51	-863.03	1802.05	0.001	0.10
$p^{\mathbb{k}}_{u}$	-868.04	1812.07	-865.80	1807.61	0.009	0.39
$\theta_{P,u}$	-878.32	1818.64	-875.87	1813.75	0.23	0.14
ξ_u	-880.84	1823.67	-879.24	1820.48	0.92	0.98
$\theta^{\mathbb{k}}_{S,u}$	-881.09	1838.18	-872.08	1820.17	0.26	0.44
$f^{\mathbb{k}}_{S,u}$	-881.53	1839.05	-872.49	1820.99	0.31	0.97
$r^{\mathbb{k}}_{u}$	-883.93	1843.87	-870.46	1816.93	0.12	0.95

TABLE S3

Comparison of model fit and complexity across various configurations of unit-specific parameters within the panelPOMP framework for mixed species Parasitized dynamics. The unit-specific parameter setting assessed include ($\theta^{\mathbb{k}}_{I,u}$), ($p^{\mathbb{k}}_{u}$), ξ_u, ($f^{\mathbb{k}}_{S,u}$), ($\theta^{\mathbb{k}}_{S,u}$), $\theta_{P,u}$ and ($r^{\mathbb{k}}_{u}$) for $\mathbb{k} \in \{\mathtt{m}, \mathtt{i}\}$.

For example, in the fourteen-parameter alternatives for testing whether θ_I is unit-specific (38 parameters in unit-specific model and 24 parameters in all-shared model) the simulated log-likelihood ratio statistic does not sit near the χ^2_{14} mean of 14: its average across these alternatives is about 27, ranging from roughly 16 for θ_S to 36 for r. This excess arises because maximizing over many added unit-specific parameters by stochastic search absorbs Monte Carlo and finite-sample fluctuation that an exactly maximized likelihood would not, and it grows with how weakly the added parameters are identified. Measured against its own bootstrap null, which is centred near 23, the observed $\Lambda = 35$ for θ_I is unremarkable, even though it is extreme against χ^2_{14}.

This violation of the asymptotic chi-square null distribution is not evident for all parameters. For example, consider adding unit-specific variation to θ_P, which adds only seven parameters. At this lower dimension the two references are in closer agreement, and both give a clearly non-significant conclusion ($p_{\chi^2} = 0.23$ against $p_{\text{boot}} = 0.14$). We accordingly take the bootstrap as the primary test and report the χ^2 values as a familiar but, in this regime, anti-conservative benchmark. One caveat concerns θ_I, the closest of the seven alternatives to significance: at $B = 100$ its bootstrap p-value of 0.10 carries a Monte Carlo standard error of order 0.03 (computed as $\sqrt{p(1-p)/B}$). It nonetheless sits well above the conventional 0.05 level, so for this panel the conclusion that no unit-specific elaboration improves on the all-shared model is supported by the data.

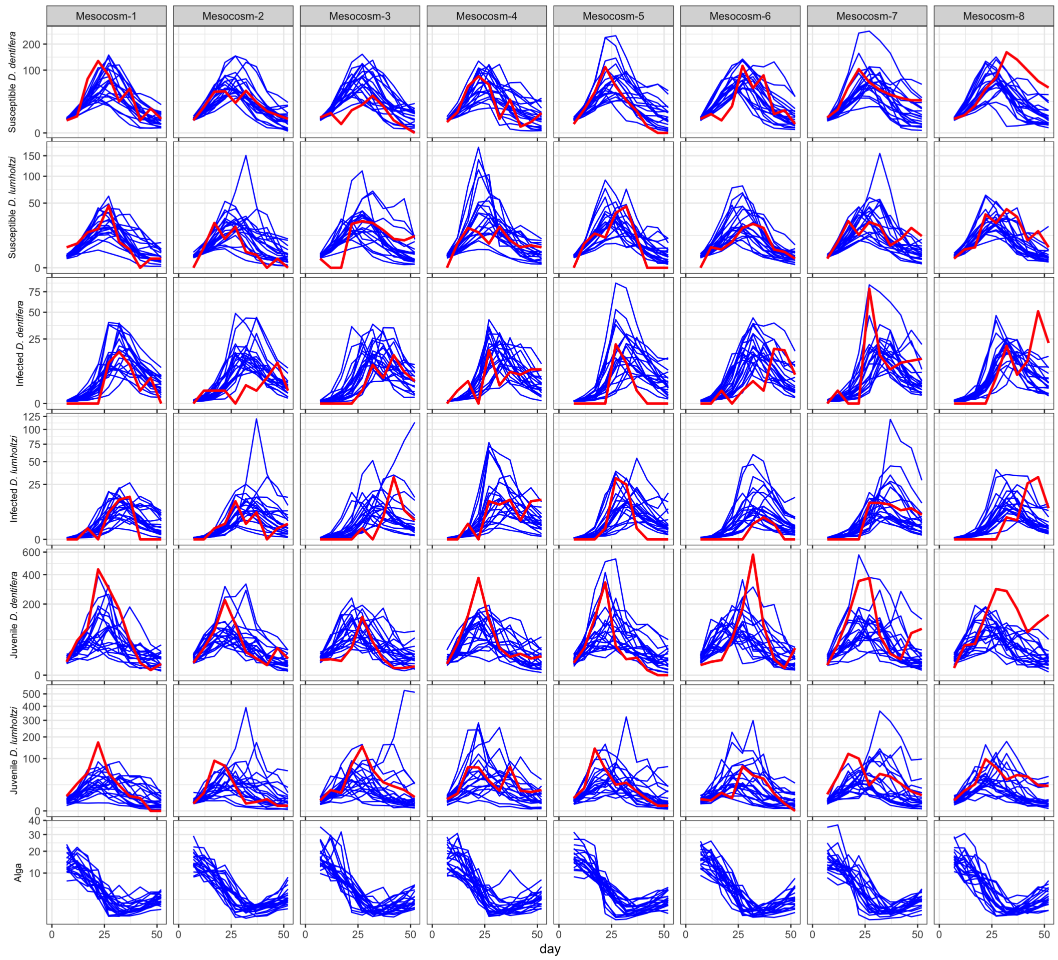
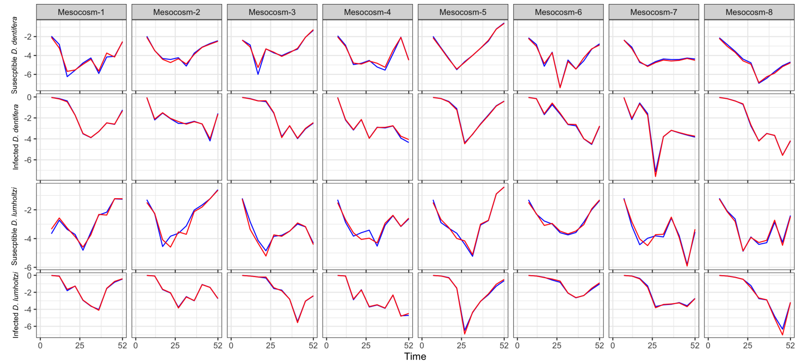
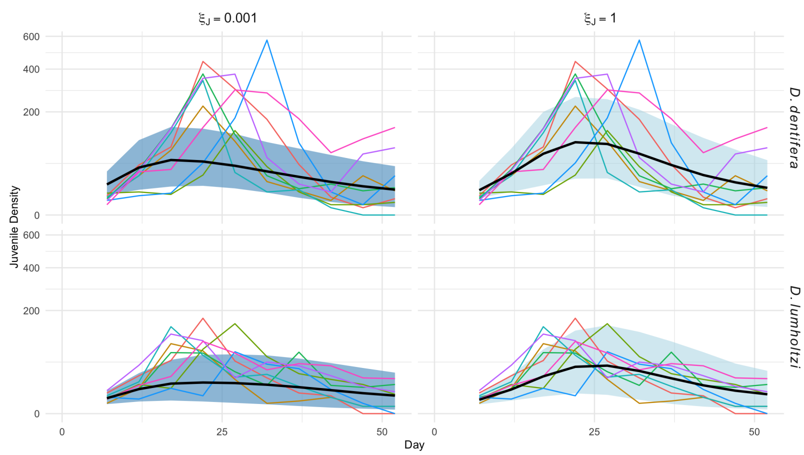


Figure S4: Simulated densities (Individuals/Liter) of susceptible, infected and juvenile *D. dentifera*, susceptible, infected and juvenile *D. lumholtzi* and alga density ($10^6\cdot$cells/Liter) over time for each experimental unit. The blue lines represent individual simulation runs, capturing the variability in susceptible density across replicates, while the red line represents the actual experiment data.



(a) Comparison of conditional log likelihood of model fitting on susceptible, infected *D. dentifera* and *D. lumholtzi* at each observation for setting of $\xi_{J,u} = 1$ (blue) and $\xi_{J,u} = 0.001$ (red).



(b) Simulated densities (Individuals/Liter) of juvenile *D. lumholtzi* and *D. dentifera* over time for in the mixed species parasitized dynamics based on maximized loglikelihood parameters, with simulated mean curves (solid black curves) and 95% confidence band for setting $\xi_{J,u} = 0.001$ (purple) and $\xi_{J,u} = 1$ (blue). The observed densities are represented by the solid colorful curves.

Figure S5: Residual analysis results

We assessed the identifiability of the juvenile filtration ratio $\xi_{J,u}$ by refitting the SIRJPF2 model with $\xi_{J,u}$ fixed at 1 (juveniles and adults filter at the same rate) and at the extreme value 0.001 (juveniles filter much more slowly). Fixing $\xi_{J,u} = 1$ gives a maximized log-likelihood of -881.13, only slightly higher than the -882.24 obtained at $\xi_{J,u} = 0.001$, so the parameter is weakly identified; the conditional log-likelihoods for susceptible and infected *D. dentifera* and *D. lumholtzi* are similar between the two settings (Figure S5a). At the biologically implausible small value the simulated trajectories contain far fewer juveniles than were observed (Figure S5b), which would bias the estimates of $r_u^{\mathbb{k}}$, $p_u^{\mathbb{k}}$ and $f_{S,u}^{\mathbb{k}}$.

S7. SIRJPF: A Single *Daphnia* Species with Parasites. The SIRJPF model presented here encapsulates the dynamics in each bucket(u) among susceptible individuals ($S^{\mathbb{k}}$), infected individuals ($I^{\Bbbk}$), juvenile population ($J^{\Bbbk}$), algae as a food resource (F), the parasite population (P). The superscript $\Bbbk$ denotes the species, which represents native or invasive species. Each stochastic differential equation accounts for various biological and ecological processes, augmented with stochastic terms to capture the inherent randomness within the systems.

$$\text{(S1)} \quad dS_u^{\Bbbk}(t) = \lambda_{J,u}^{\Bbbk} J_u^{\Bbbk}(t)\,dt - \left\{\theta_{S,u}^{\Bbbk} + p_u^{\Bbbk} f_{S,u}^{\Bbbk} P_u(t) + \delta\right\} S_u^{\Bbbk}(t)\,dt + S_u^{\Bbbk}(t)\,d\zeta_{S,u}^{\Bbbk},$$

$$\text{(S2)} \quad dJ_u^{\Bbbk}(t) = r_u^{\Bbbk} f_{S,u}^{\Bbbk} F_u(t) S_u^{\Bbbk}(t)\,dt - \theta_{J,u}^{\Bbbk} J_u^{\Bbbk}(t)\,dt - \delta J_u^{\Bbbk}(t)\,dt - \lambda_{J,u}^{\Bbbk} J_u^{\Bbbk}(t)\,dt + J_u^{\Bbbk}(t)\,d\zeta_{J,u}^{\Bbbk},$$

$$\text{(S3)} \quad dI_u^{\Bbbk}(t) = p_u^{\Bbbk} f_{S,u}^{\Bbbk} S_u^{\Bbbk}(t) P_u(t)\,dt - \left\{\theta_{I,u}^{\Bbbk} + \delta\right\} I_u^{\Bbbk}(t)\,dt + I_u^{\Bbbk}(t)\,d\zeta_{I,u}^{\Bbbk},$$

$$\text{(S4)} \quad dP_u(t) = \beta_u^{\Bbbk} \theta_{I,u}^{\Bbbk} I_u^{\Bbbk}(t) dt - f_{S,u}^{\Bbbk}(S_u^{\Bbbk}(t) + \xi_u I_u^{\Bbbk}(t)) P_u(t) dt - \theta_{p,u} P_u(t)\,dt + P_u(t)\,d\zeta_{P,u},$$

$$\text{(S5)} \quad dF_u(t) = -f_{S,u}^{\Bbbk} F_u(t) \left(S_u^{\Bbbk}(t) + \xi_{J,u} J_u^{\Bbbk}(t) + \xi_u I_u^{\Bbbk}(t)\right) dt + \mu\,dt + F_u(t)\,d\zeta_{F,u},$$

$$\text{(S6)} \quad d\zeta_{S,u}^{\Bbbk} \sim \mathrm{N}\big[0, (\sigma_{S,u}^{\Bbbk})^2\,dt\big], d\zeta_{I,u}^{\Bbbk} \sim \mathrm{N}\big[0, (\sigma_{I,u}^{\Bbbk})^2\,dt\big], d\zeta_{J,u}^{\Bbbk} \sim \mathrm{N}\big[0, (\sigma_{J,u}^{\Bbbk})^2\,dt\big],$$

$$\text{(S7)} \quad d\zeta_{F,u} \sim \mathrm{N}\big[0, (\sigma_{F,u})^2\,dt\big], d\zeta_{P,u} \sim \mathrm{N}\big[0, (\sigma_{P,u})^2\,dt\big].$$

Each equation represents specific biological interactions and processes. Equation (S1) models the dynamics of the susceptible *Daphnia* population across time, $S_u^{\Bbbk}(t)$, where growth occurs from the juvenile population $J_u^{\Bbbk}(t)$ at a maturation rate $\lambda_{J,u}^{\Bbbk}$, while losses include natural mortality ($\theta_{S,u}^{\Bbbk}$), infection by intaking parasite at rate $p_u^{\Bbbk} f_{S,u}^{\Bbbk} P_u^{\Bbbk}(t)$, and sampling rate ($\delta$).

In Equation (S2), the dynamics of the juvenile population $J_u^{\Bbbk}(t)$ are captured, with growth occurring via feeding on algae, scaled by birth rate $r_u^{\Bbbk}$ and feeding rate $f_{S,u}^{\Bbbk} F_u^{\Bbbk}(t)$. Losses in $J_u^{\Bbbk}(t)$ include natural mortality ($\theta_{J,u}^{\Bbbk}$), sampling rate (δ), and maturation to the adult population at rate $\lambda_{J,u}^{\Bbbk}$. Additionally, $J_u^{\Bbbk}(t)\,d\zeta_{J,u}^{\Bbbk}$ represents stochastic fluctuations affecting the juvenile population.

Equation (S3) describes the infected population $I_u^{\Bbbk}(t)$, with new infections arising from interaction between susceptible individuals $S_u^{\Bbbk}(t)$ and alga at rate $p_u^{\Bbbk} f_{S,u}^{\Bbbk} S_u^{\Bbbk}(t) P_u^{\Bbbk}(t)$. Infected individuals experience losses through natural mortality ($\theta_{I,u}^{\Bbbk}$) and sampling rate (δ), while $I_u^{\Bbbk}(t)\,d\zeta_{I,u}^{\Bbbk}$ accounts for stochastic noise.

Equation (S4) models the dynamics of *M.* population $P_u(t)$, which grows through spore release from the death of infected individuals $I_u^{\Bbbk}(t)$ at a rate $\beta_u^{\Bbbk}\theta_{I,u}^{\Bbbk}$. Losses are due to intaking by susceptible and infected individuals, scaled by filtering rate $f_{S,u}^{\Bbbk}$ and the ratio of filtering rate ξ_u between subjects before and after infection, and natural mortality at rate $\theta_{P,u}^{\Bbbk}$. Stochastic fluctuations are represented by the term $P_u(t)\,d\zeta_{P,u}$.

Finally, Equation (S5) models the *A.* alga population $F_u(t)$. The algae are consumed by susceptible individuals, juveniles, and infected individuals, scaled by consumption rate f_S and ratio of filtering rate $\xi_{J,u}$ and ξ_u. Algae gets manually refilled at rate μ, with $F_u(t)\,d\zeta_{F,u}$ representing stochastic noise.

The stochastic terms $d\zeta_{S,u}^{\Bbbk}$, $d\zeta_{J,u}^{\Bbbk}$, $d\zeta_{I,u}^{\Bbbk}$, $d\zeta_{P,u}$, and $d\zeta_{F,u}$ represent Brownian motion affecting each population, modeled as Gaussian white noise with mean zero and variances $(\sigma_{S,u}^{\Bbbk})^2\,dt$, $(\sigma_{J,u}^{\Bbbk})^2\,dt$, $(\sigma_{I,u}^{\Bbbk})^2\,dt$, $(\sigma_{P,u})^2\,dt$, and $(\sigma_{F,u})^2\,dt$, respectively. Similar to the SIR-JPF2 model, we fixed the $\sigma_S^{\Bbbk}$ to be zero due to its flat profile plot.

This model describes the interactions and feedback mechanisms among the populations under study. The equations describe the growth or decline of each population, influenced by deterministic factors such as reproduction, filtering rates, predation, infection, and mortality, alongside stochastic factors captured through the $d\zeta_{\cdot,u}$ terms for each unit. These stochastic components represent environmental variability and other unpredictable influences, adhering to the principles of stochastic differential equations. By incorporating random fluctuations, the model offers a statistically motivated representation of population dynamics, extending beyond the deterministic framework used in previous analysis. In the following sections, we will present the flow diagram and parameter estimation results to further illustrate the interaction between latent states.

The SIRJPF model illustrates the interdependent dynamics among the Susceptible (S), Infected (I), Juvenile (J), Food (F), and Parasite (P) populations, with transitions representing ecological and biological interactions. The model integrates deterministic growth and mortality rates with stochastic components, providing a realistic framework for studying population fluctuations under variable environmental conditions.

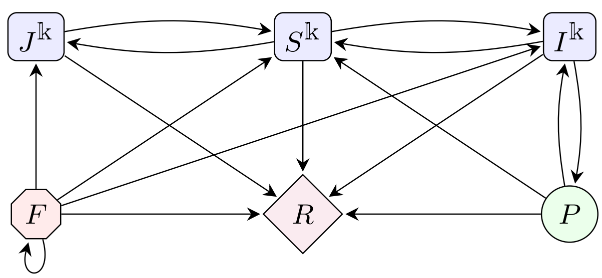


Figure S6: Flow diagram for the SIRJPF model illustrating population interactions. The model includes the R state, representing mortality. Susceptible populations ($S^{\Bbbk}$) can reproduce into juvenile ($J^{\Bbbk}$), where the superscript $\Bbbk$ denotes the species of *Daphnia*. And *A*. Algae (F) will be refilled, as shown by the recycling arrows. Transitions from $S^{\Bbbk}$ to $I^{\Bbbk}$ denote infection, and transitions to the R state occur upon death, where deceased individuals release parasites back into the environment. All $S^{\Bbbk}$, $I^{\Bbbk}$ and $J^{\Bbbk}$ consume resources from F, and over time, components in F and P also progress to R.

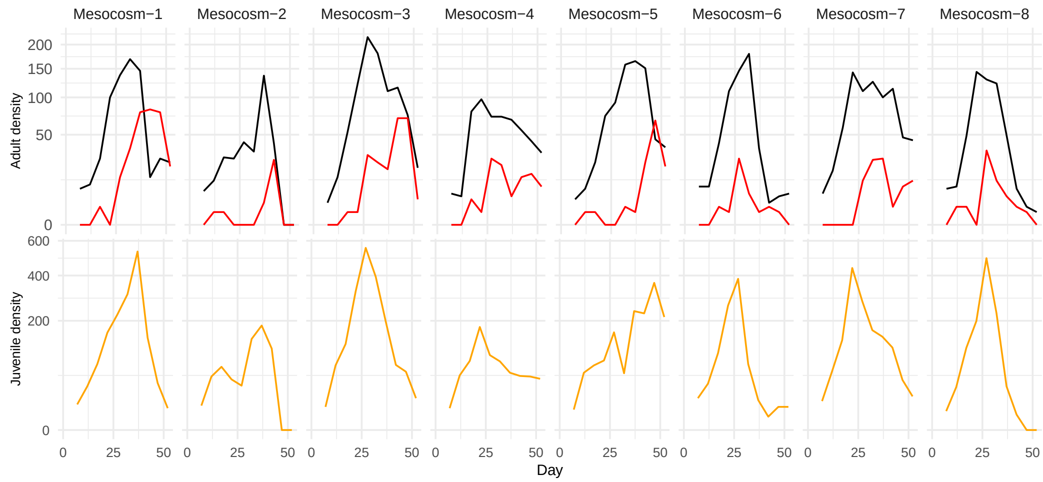


Figure S7: Density (Individuals/Liter) of *D. dentifera*. The top panel shows adult susceptibles (*D. dentifera*, black) and infecteds (*D. dentifera*, red). The bottom panel shows juvenile susceptibles (*D. dentifera*, orange). There were negligible infected juveniles. Columns are buckets corresponding to replications with same treatment setting.

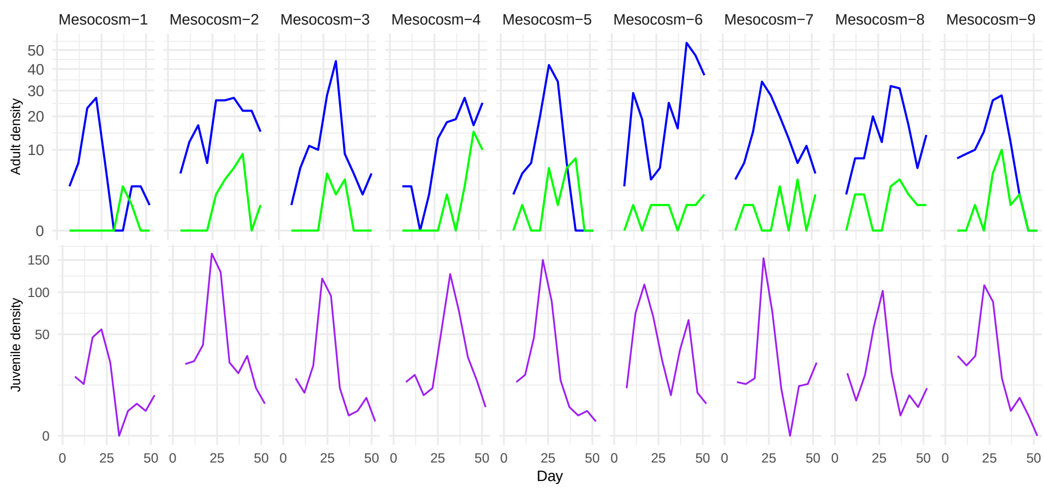


Figure S8: Density (Individuals/Liter) of *D. lumholtzi*. The top panel shows adult susceptibles (*D. lumholtzi*, blue) and infecteds (*D. lumholtzi*, green). The bottom panel shows juvenile susceptibles (*D. lumholtzi*, purple). There were negligible infected juveniles. Columns are buckets corresponding to replications with same treatment setting.

Parameter	Definition	Unit	Value	CI
$S^{\mathbb{n}}$	Susceptible host density	$\text{individual} \cdot L^{-1}$	Variable	
$I^{\mathbb{n}}$	Infected host density	$\text{individual} \cdot L^{-1}$	Variable	
$J^{\mathbb{n}}$	Juvenile host density	$\text{individual} \cdot L^{-1}$	Variable	
F	Alga density	$10^6 \cdot \text{cell} \cdot L^{-1}$	Variable	
P	Spore density	$10^3 \cdot \text{spore} \cdot L^{-1}$	Variable	
$r^{\mathbb{n}}$	Birth rate of juvenile	$\text{individual} \cdot 10^{-6} \cdot \text{cell}^{-1}$	$5.52 \cdot 10$	$(3.67 \cdot 10, 1.68 \cdot 10^2)$
$f_S^{\mathbb{n}}$	Susceptible adult host filtering rate	$\text{L} \cdot \text{individual}^{-1} \cdot \text{day}^{-1}$	$7.64 \cdot 10^{-4}$	$(2.56 \cdot 10^{-4}, 1.31 \cdot 10^{-3})$
$p^{\mathbb{n}}$	Number of infections per spore	$10^{-3} \cdot \text{individual} \cdot \text{spore}^{-1}$	$3.06 \cdot 10^{-1}$	$(1.63 \cdot 10^{-1}, 8.67 \cdot 10^{-1})$
$p^{\mathbb{n}} \cdot f_{S^{\mathbb{n}}}$	Effective infection rate of native adult hosts per-spore	$10^{-3} \cdot \text{L} \cdot \text{spore}^{-1} \cdot \text{day}^{-1}$	$2.34 \cdot 10^{-4}$	$(1.77 \cdot 10^{-4}, 3.09 \cdot 10^{-4})$
$r^{\mathbb{n}} \cdot f_{S^{\mathbb{n}}}$	Effective Birth rate of native juvenile	$10^{-6} \cdot \text{L} \cdot \text{cell}^{-1} \cdot \text{day}^{-1}$	$4.22 \cdot 10^{-2}$	$(3.81 \cdot 10^{-2}, 5.19 \cdot 10^{-2})$
$\theta_S^{\mathbb{n}}$	Susceptible adult host mortality rate	day^{-1}	$9.30 \cdot 10^{-7}$	$(0, 1.83 \cdot 10^{-4})$
$\theta_I^{\mathbb{n}}$	Infected adult host mortality rate	day^{-1}	$4.54 \cdot 10^{-1}$	$(2.03 \cdot 10^{-1}, 7.57 \cdot 10^{-1})$
$\theta_J^{\mathbb{n}}$	Juvenile mortality rate	day^{-1}	$7.69 \cdot 10^{-5}$	$(0, 4.43 \cdot 10^{-2})$
θ_P	Spore degradation rate	day^{-1}	$1.18 \cdot 10^{-4}$	$(0, 2.15 \cdot 10^{-2})$
$\lambda_J^{\mathbb{n}}$	Maturation rate of the juvenile	day^{-1}	$1.00 \cdot 10^{-1}$	
ξ	Ratio of infected host filtering rate	Unitless	$1.13 \cdot 10$	$(2.48, 4.37 \cdot 10)$
ξ_J	Ratio of juvenile individual filtering rate	Unitless	1.00	
$\beta^{\mathbb{n}}$	Spores produced by death per infected individual	$10^3 \cdot \text{spore} \cdot \text{individual}^{-1} \cdot \text{day}^{-1}$	$3.00 \cdot 10$	
μ	Algae refilling rate	$10^6 \cdot \text{cell} \cdot \text{L}^{-1} \cdot \text{day}^{-1}$	$3.70 \cdot 10^{-1}$	
δ	Sampling rate	day^{-1}	$1.30 \cdot 10^{-2}$	
$\sigma_S^{\mathbb{n}}$	Standard deviation of Brownian motion of susceptible adult	$\sqrt{\text{individual} \cdot \text{day}^{-1}}$	0	
$\sigma_I^{\mathbb{n}}$	Standard deviation of Brownian motion of infected adult	$\sqrt{\text{individual} \cdot \text{day}^{-1}}$	$5.74 \cdot 10^{-1}$	$(0, 6.19 \cdot 10^{-1})$
$\sigma_J^{\mathbb{n}}$	Standard deviation of Brownian motion of juvenile	$\sqrt{\text{individual} \cdot \text{day}^{-1}}$	$3.41 \cdot 10^{-1}$	$(1.52 \cdot 10^{-1}, 5.18 \cdot 10^{-1})$
σ_F	Standard deviation of Brownian motion of alga	$\sqrt{\text{individual} \cdot \text{day}^{-1}}$	$6.97 \cdot 10^{-2}$	$(0, 1.63 \cdot 10^{-1})$
σ_P	Standard deviation of Brownian motion of parasite	$\sqrt{\text{individual} \cdot \text{day}^{-1}}$	$4.60 \cdot 10^{-1}$	$(2.10 \cdot 10^{-1}, 6.10 \cdot 10^{-1})$
$\tau_S^{\mathbb{n}}$	Measurement dispersion for susceptible adult	Unitless	$1.50 \cdot 10$	$(7.25, 3.99 \cdot 10)$
$\tau_I^{\mathbb{n}}$	Measurement dispersion for infected adult	Unitless	1.13	$(6.80 \cdot 10^{-1}, 2.61)$

TABLE S4

Variables and parameter definitions and estimates of SIRJPF model for D. dentifera-only dynamics. The confidence interval is generated by the Monte Carlo Adjusted Profile.

Parameter	Definition	Unit	Value	CI
$S^{\mathbb{i}}$	Susceptible host density	$\text{individual} \cdot L^{-1}$	Variable	
$I^{\mathbb{i}}$	Infected host density	$\text{individual} \cdot L^{-1}$	Variable	
$J^{\mathbb{i}}$	Juvenile host density	$\text{individual} \cdot L^{-1}$	Variable	
F	Alga density	$10^6 \cdot \text{cell} \cdot L^{-1}$	Variable	
P	Spore density	$10^3 \cdot \text{spore} \cdot L^{-1}$	Variable	
$r^{\mathbb{i}}$	Birth rate of juvenile	$\text{individual} \cdot 10^{-6} \cdot \text{cell}^{-1}$	$2.49 \cdot 10^2$	$(9.36 \cdot 10, 1.63 \cdot 10^3)$
$f_S^{\mathbb{i}}$	Susceptible adult host filtering rate	$\text{L} \cdot \text{individual}^{-1} \cdot \text{day}^{-1}$	$6.49 \cdot 10^{-4}$	$(6.57 \cdot 10^{-5}, 1.33 \cdot 10^{-3})$
$p^{\mathbb{i}}$	Number of infections per spore	$10^{-3} \cdot \text{individual} \cdot \text{spore}^{-1}$	$6.65 \cdot 10^{-1}$	$(3.42 \cdot 10^{-1}, 6.17)$
$p^{\mathbb{i}} \cdot f_{S^{\mathbb{i}}}$	Effective infection rate of invasive adult hosts per-spore	$10^{-3} \cdot \text{L} \cdot \text{spore}^{-1} \cdot \text{day}^{-1}$	$4.32 \cdot 10^{-4}$	$(3.78 \cdot 10^{-4}, 7.45 \cdot 10^{-4})$
$r^{\mathbb{i}} \cdot f_{S^{\mathbb{i}}}$	Effective Birth rate of invasive juvenile	$10^{-6} \cdot \text{L} \cdot \text{cell}^{-1} \cdot \text{day}^{-1}$	$1.62 \cdot 10^{-1}$	$(1.08 \cdot 10^{-1}, 2.48 \cdot 10^{-1})$
$\theta_S^{\mathbb{i}}$	Susceptible adult host mortality rate	day^{-1}	$5.73 \cdot 10^{-1}$	$(3.01 \cdot 10^{-1}, 1.16)$
$\theta_I^{\mathbb{i}}$	Infected adult host mortality rate	day^{-1}	$1.74 \cdot 10^{-1}$	$(2.05 \cdot 10^{-2}, 4.16 \cdot 10^{-1})$
$\theta_J^{\mathbb{i}}$	Juvenile mortality rate	day^{-1}	$2.55 \cdot 10^{-5}$	$(0, 1.28 \cdot 10^{-1})$
θ_P	Spore degradation rate	day^{-1}	$1.71 \cdot 10^{-6}$	$(0, 8.61 \cdot 10^{-2})$
$\lambda_J^{\mathbb{i}}$	Maturation rate of the juvenile	day^{-1}	$1.00 \cdot 10^{-1}$	
ξ	Ratio of infected host filtering rate	Unitless	$7.33 \cdot 10$	$(9.54, 8.02 \cdot 10^2)$
ξ_J	Ratio of juvenile individual filtering rate	Unitless	1.00	
$\beta^{\mathbb{i}}$	Spores produced by death per infected individual	$10^3 \cdot \text{spore} \cdot \text{individual}^{-1} \cdot \text{day}^{-1}$	$3.00 \cdot 10$	
μ	Algae refilling rate	$10^6 \cdot \text{cell} \cdot \text{L}^{-1} \cdot \text{day}^{-1}$	$3.70 \cdot 10^{-1}$	
δ	Sampling rate	day^{-1}	$1.30 \cdot 10^{-2}$	
$\sigma_S^{\mathbb{i}}$	Standard deviation of Brownian motion of susceptible adult	$\sqrt{\text{individual} \cdot \text{day}^{-1}}$	0	
$\sigma_I^{\mathbb{i}}$	Standard deviation of Brownian motion of infected adult	$\sqrt{\text{individual} \cdot \text{day}^{-1}}$	$2.73 \cdot 10^{-1}$	$(0, 5.75 \cdot 10^{-1})$
$\sigma_J^{\mathbb{i}}$	Standard deviation of Brownian motion of juvenile	$\sqrt{\text{individual} \cdot \text{day}^{-1}}$	$3.72 \cdot 10^{-1}$	$(2.75 \cdot 10^{-1}, 5.25 \cdot 10^{-1})$
σ_F	Standard deviation of Brownian motion of alga	$\sqrt{\text{individual} \cdot \text{day}^{-1}}$	$5.22 \cdot 10^{-8}$	$(0, 1.80 \cdot 10^{-2})$
σ_P	Standard deviation of Brownian motion of parasite	$\sqrt{\text{individual} \cdot \text{day}^{-1}}$	$7.99 \cdot 10^{-8}$	$(0, 3.55 \cdot 10^{-1})$
$\tau_S^{\mathbb{i}}$	Measurement dispersion for susceptible adult	Unitless	$7.00 \cdot 10$	$(0, 4.40 \cdot 10^3)$
$\tau_I^{\mathbb{i}}$	Measurement dispersion for infected adult	Unitless	1.17	$(5.80 \cdot 10^{-1}, 1.08 \cdot 10)$

TABLE S5

Variables and parameter definitions and estimates of SIRJPF model for D. lumholtzi-only dynamics. The confidence interval is generated by the Monte Carlo Adjusted Profile.

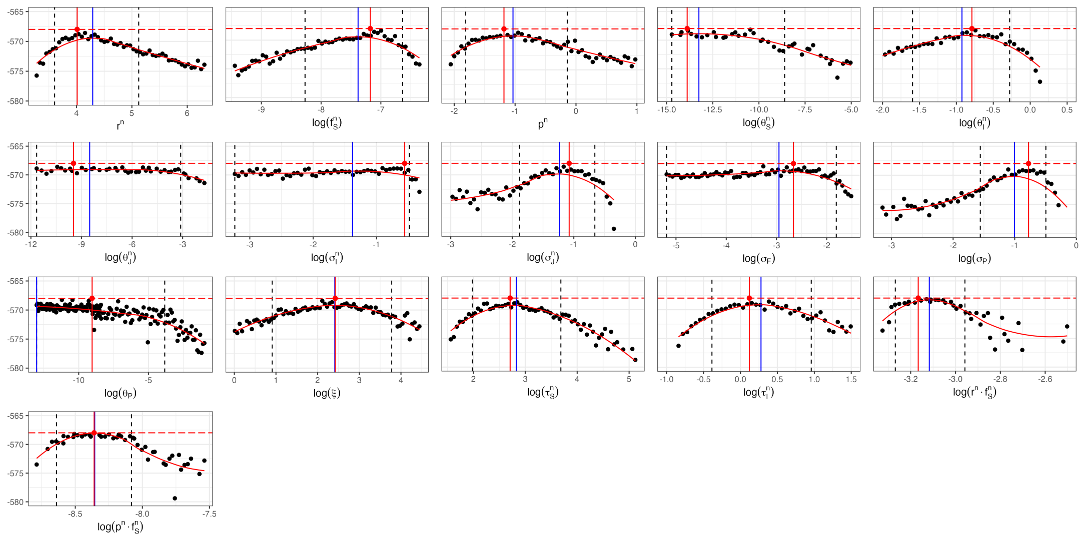


Figure S9: Monte Carlo Adjusted Profile results of SIRJPF model for *D. dentifera*-only dynamics. The vertical dotted lines represent the 95% confidence interval obtained by MCAP. The vertical blue lines show the MLE estimated using MCAP. The red vertical lines correspond to the model with the overall highest likelihood among all searches.

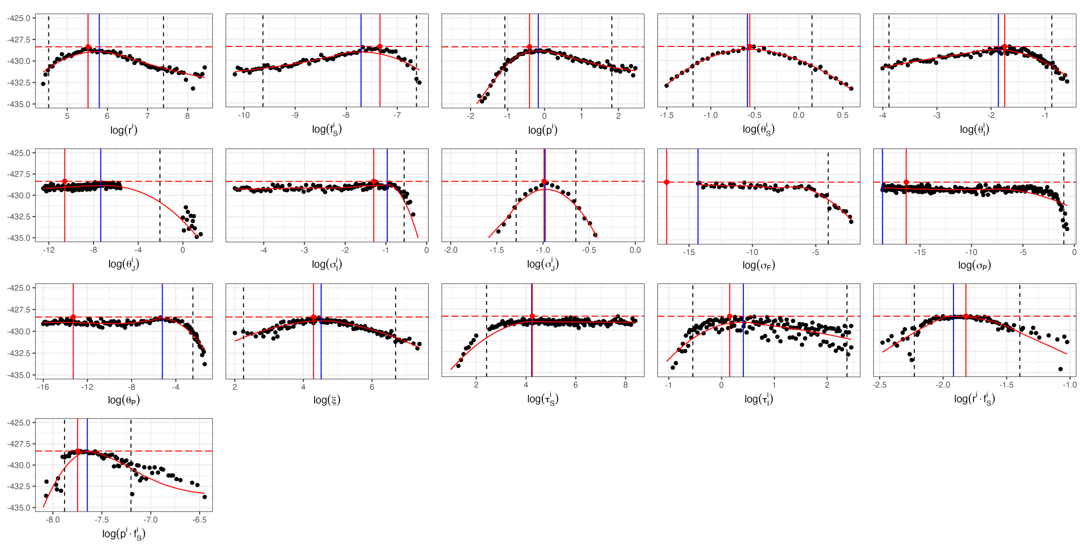


Figure S10: Monte Carlo Adjusted Profile results of SIRJPF model for *D. lumholtzi*-only dynamics. The vertical dotted lines represent the 95% confidence interval obtained by MCAP. The vertical blue lines show the MLE estimated using MCAP. The red vertical lines correspond to the model with the overall highest likelihood among all searches.

S7.1. *Results.* For the invasive *D. lumholtzi* treatment (Table S7), the unit-specific ξ_u model attains the lowest AIC, but the improvement over the all-shared model is marginal and the bootstrap likelihood ratio test is not significant ($p_{\text{boot}} = 0.07$), so we adopt the all-shared model. For the native *D. dentifera* treatment (Table S6), a unit-specific infected-host

Specific parameters	Max loglik (PIF)	AIC (PIF)	Max loglik (MPIF)	AIC (MPIF)	LRT p-value (χ^2)	LRT p-value (bootstrap)
$\theta^{\mathbb{m}}_{I,u}$	-549.38	1140.77	-547.92	1137.84	<0.001	0.010
$\varnothing$	-567.98	1163.97	-567.98	1163.97	—	—
$r^{\mathbb{m}}_{u}$	-563.38	1168.76	-562.95	1167.90	0.18	0.85
$p^{\mathbb{m}}_{u}$	-563.78	1169.56	-562.75	1167.49	0.16	0.79
ξ_u	-566.59	1175.19	-565.80	1173.60	0.74	0.94
$\theta^{\mathbb{m}}_{S,u}$	-567.31	1176.61	-565.48	1172.96	0.66	0.87
$f^{\mathbb{m}}_{S,u}$	-568.66	1179.33	-566.32	1174.63	0.85	>0.99
$\theta_{P,u}$	-568.21	1178.42	-568.87	1179.74	>0.99	0.94

TABLE S6

Comparison of model fit and complexity across various configurations of unit-specific parameters within the panelPOMP framework for D. dentifera-only dynamics. The unit-specific parameter setting assessed include $\theta^{\mathbb{m}}_{I,u}$, $p^{\mathbb{m}}_{u}$, $r^{\mathbb{m}}_{u}$, ξ_u, $\theta^{\mathbb{m}}_{S,u}$, $\theta_{P,u}$.

Specific parameters	Max loglik (PIF)	AIC (PIF)	Max loglik (MPIF)	AIC (MPIF)	LRT p-value (χ^2)	LRT p-value (bootstrap)
ξ_u	-419.78	883.560	-416.87	877.747	0.003	0.07
$\varnothing$	-428.36	884.715	-428.36	884.715	—	—
$p^{\mathbb{i}}_{u}$	-421.21	886.411	-420.66	885.326	0.05	0.31
$\theta^{\mathbb{i}}_{I,u}$	-421.21	886.427	-419.32	882.636	0.02	0.15
$f^{\mathbb{i}}_{S,u}$	-422.85	889.700	-423.26	890.523	0.25	0.60
$\theta_{P,u}$	-429.20	902.400	-428.99	901.972	>0.99	0.84
$r^{\mathbb{i}}_{u}$	-432.43	908.868	-426.31	896.612	0.85	0.91
$\theta^{\mathbb{i}}_{S,u}$	-432.87	909.740	-430.01	904.014	>0.99	>0.99

TABLE S7

Comparison of model fit and complexity across various configurations of unit-specific parameters within the panelPOMP framework for D. lumholtzi-only dynamics. The unit-specific parameter setting assessed include $\theta^{\mathbb{i}}_{I,u}$, $p^{\mathbb{i}}_{u}$, $r^{\mathbb{i}}_{u}$, ξ_u, $\theta^{\mathbb{i}}_{S,u}$, $\theta_{P,u}$.

mortality rate $\theta^{\mathbb{m}}_{I,u}$ attains a substantially lower AIC ($\Delta\text{AIC} \approx 26$) and is favored by the bootstrap test ($p_{\text{boot}} = 0.010$). This is the only unit-specific alternative to reject the all-shared null among the 30 tested across the six treatment-by-species combinations, and we interpret it with caution. Infected-host mortality is the parameter most sensitive to variation in the parasite challenge that is difficult to control across replicate mesocosms, such as the spore dose that becomes established and the within-host infection intensity that follows. A mesocosm-specific $\theta^{\mathbb{m}}_{I,u}$ most plausibly absorbs this replicate-level variation in the infection process, rather than indicating that the underlying population dynamics differ between mesocosms. The heterogeneity is confined to a single infection-outcome parameter in a single treatment, whereas every other parameter and every other treatment is adequately described by shared dynamics with process noise.

We therefore adopt the all-shared model as the description of the shared biological dynamics, and regard the *D. dentifera* $\theta^{\mathbb{m}}_{I,u}$ heterogeneity as a localized feature of the infection process. A subsequent experiment that controls the initial parasite exposure could investigate this directly.

S7.2. *Simulation.* The simulation plots (Figures S11 to S14) demonstrate our model's capability to capture the complex interactions among various states within the population dynamics of *D. dentifera* and *D. lumholtzi*. Specifically, the model captures the latent pattern for susceptible and infected densities across multiple experimental units, as evidenced by the close alignment between the simulated mean trajectories (depicted by the black and light-blue bands) and the observed experimental data (colorful solid lines). Each blue line represents an individual simulation case, capturing the stochastic variability across replicates, while the light blue shaded area (where present) denotes the 95% confidence interval of the simulation results. Remarkably, although juvenile density data were not included in the model fitting process, the model still effectively predicts juvenile density trends. The promising performance of the model indicates the model's capacity to generalize underlying ecological processes, as the 95% confidence intervals largely cover the observed juvenile data.

At the same time, the depletion of algal density over time, reflected in the simulation further explains the behavior of the *Daphnia*. The decline in algal density at later stages indicates that the system reaches a state of food limitation, which is a dynamic anticipated by our model. The reduction in food availability provides a mechanistic explanation for the observed trends in *Daphnia* densities, as resource scarcity constrains population growth, ultimately affecting the susceptible, infected, and juvenile densities. Thus, the model captures not only the reproductive dynamics but also the resource-consumption interactions that shape these dynamics. The model's ability to reproduce these patterns accurately highlights its potential utility in explaining and predicting empirical trends in complex ecological systems.

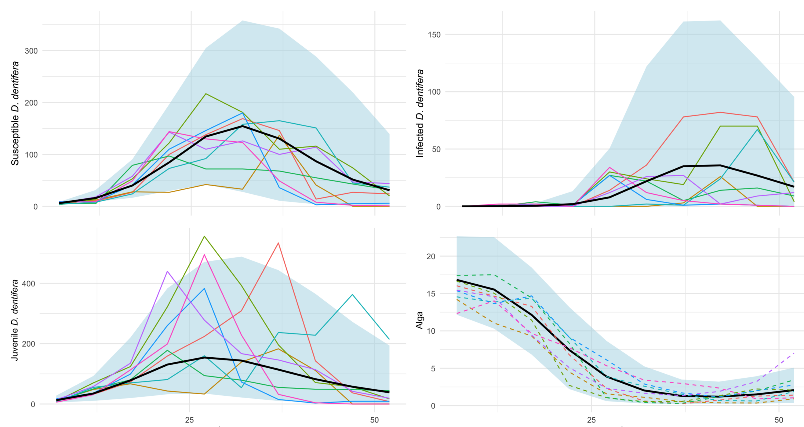


Figure S11: Simulation plots of densities (Individuals/Liter) for susceptible, infected, and juvenile *D. dentifera*, along with alga density (10^6·cells/Liter) and parasite density (10^3·spores/Liter), over time (days). The colorful solid lines represent observed experimental data, showing the variability across different experimental replicates. The black line represents the mean trajectory from the simulation model, capturing the general trend of each density. The light blue shaded area corresponds to the 95% confidence interval of the simulations, indicating the range of model predictions. Dashed lines illustrate individual simulated trajectories, highlighting the model's ability to capture fluctuations around the mean trajectory.

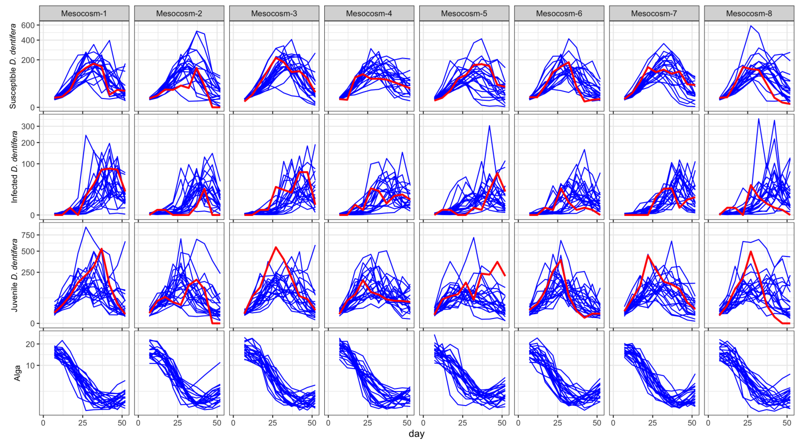


Figure S12: Simulated densities (Individuals/Liter) of susceptible *D. dentifera*, infected *D. dentifera* density, juvenile *D. dentifera* and alga density (10^6·cells/Liter) over time (days) for each experimental unit. The blue lines represent individual simulation runs, capturing the variability in susceptible density across replicates, while the red line represents the actual experimental data.

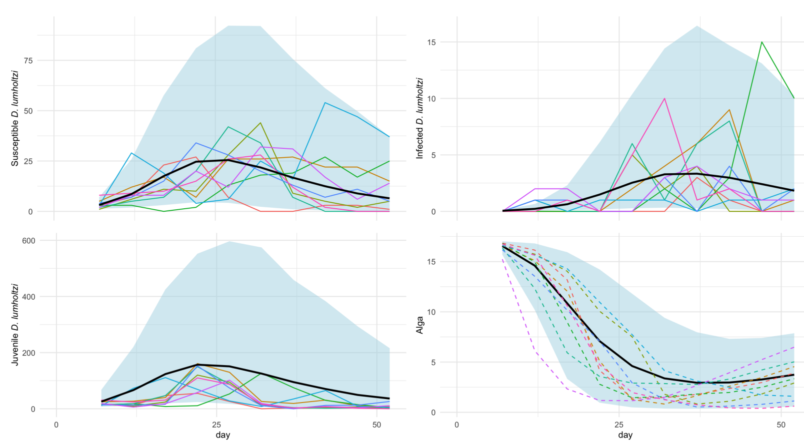


Figure S13: Simulation plots of densities (Individuals/Liter) for susceptible, infected, and juvenile *D. lumholtzi*, along with alga density (10^6·cells/Liter) and parasite density (10^3·spores/Liter), over time (days). The colorful solid lines represent observed experimental data, showing the variability across different experimental replicates. The black line represents the mean trajectory from the simulation model, capturing the general trend of each density. The light blue shaded area corresponds to the 95% confidence interval of the simulations, indicating the range of model predictions. Dashed lines illustrate individual simulated trajectories, highlighting the model's ability to capture fluctuations around the mean trajectory.

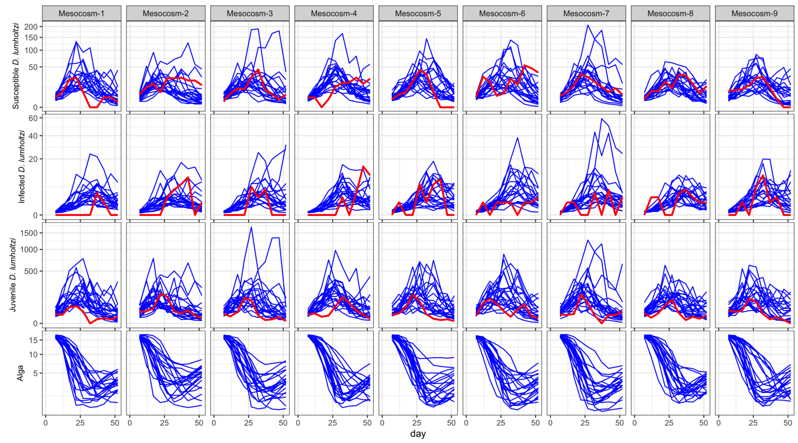


Figure S14: Simulated trajectories of susceptible *D. lumholtzi*, infected *D. lumholtzi*, juvenile *D. lumholtzi* and alga density (10^6·cells/Liter) over time (days) for each experimental unit (u1 to u9). The blue lines represent individual simulation runs, capturing the variability in susceptible density across replicates, while the red line represents the actual experimental data.

S8. SRJF2: Two *Daphnia* Species and No Parasites. In this section, we introduce the SRJF2 model, which analyze the dynamics between native and invasive species competing for a shared food resource, algae (F) in each bucket(u). The stochastic differential equations governing the system are:

$$dS_u^{\Bbbk}(t) = \lambda_{J,u}^{\Bbbk} J_u^{\Bbbk}(t)\, dt - (\theta_{S,u}^{\Bbbk} + \delta) S_u^{\Bbbk}(t)\, dt + S_u^{\Bbbk}(t)\, d\zeta_{S,u}^{\Bbbk}, \tag{S8}$$

$$dJ_u^{\Bbbk}(t) = r_u^{\Bbbk} f_{S,u}^{\Bbbk} F_u(t) S_u^{\Bbbk}(t)\, dt - \theta_{J,u}^{\Bbbk} J_u^{\Bbbk}(t)\, dt - \delta J_u^{\Bbbk}(t)\, dt - \lambda_{J,u}^{\Bbbk} J_u^{\Bbbk}(t)\, dt + J_u^{\Bbbk}(t)\, d\zeta_{J,u}^{\Bbbk}, \tag{S9}$$

$$dF_u(t) = -\sum_{\Bbbk \in \{\mathbb{n}, \mathbb{i}\}} f_{S,u}^{\Bbbk} F_u(t) \left(S_u^{\Bbbk}(t) + \xi_{J,u} J_u^{\Bbbk}(t) \right) dt + \mu\, dt + F_u(t)\, d\zeta_{F,u}, \tag{S10}$$

$$d\zeta_{S,u}^{\Bbbk} \sim \mathrm{N}\big[0, (\sigma_{S,u}^{\Bbbk})^2\, dt\big]\; d\zeta_{J,u}^{\Bbbk} \sim \mathrm{N}\big[0, (\sigma_{J,u}^{\Bbbk})^2\, dt\big]\; d\zeta_{F,u} \sim \mathrm{N}\big[0, \sigma_{F,u}^2\, dt\big]. \tag{S11}$$

Equation (S8) models the rate of change of the specific species' susceptible population $S_u^{\Bbbk}(t)$. The term $\lambda_{J,u}^{\Bbbk} J_u^{\Bbbk}(t)\, dt$ represents the maturation of juveniles into susceptible adults at rate $\lambda_{J,u}^{\Bbbk}$. The mortality term $-\left(\theta_{S,u}^{\Bbbk} + \delta\right) S_u^{\Bbbk}(t)\, dt$ accounts for natural death at rate $\theta_{S,u}^{\Bbbk}$ and the sampling rate δ. For consistency with the SIRJPF2 and SIRJPF model, we further fix $\sigma_{S,u}^{\Bbbk}$ to be zero for specific analysis of this dynamics.

Equation (S9) describes the dynamics of the juvenile population $J_u^{\Bbbk}(t)$. The growth term $r_u^{\Bbbk} f_{S,u}^{\Bbbk} F_u(t) S_u^{\Bbbk}(t)\, dt$ signifies the production of juveniles through reproduction, where $r_u^{\Bbbk}$ is the growth factor and $f_{S,u}^{\Bbbk}$ is the filtering rate of susceptible individuals on algae. The loss term $-\left(\theta_{J,u}^{\Bbbk} + \delta + \lambda_{J,u}^{\Bbbk}\right) J^{\Bbbk}(t)\, dt$ includes natural juvenile mortality at rate $\theta_{J,u}^{\Bbbk}$, sampling rate δ, and maturation into the susceptible adult class at rate $\lambda_{J,u}^{\Bbbk}$. The stochastic term $J_u^{\Bbbk}(t)\, d\zeta_{J,u}^{\Bbbk}$ captures random fluctuations affecting the juvenile population, with $d\zeta_{J,u}^{\Bbbk}$ being the Brownian motion characterized by variance $(\sigma_{J,u}^{\Bbbk})^2\, dt$.

Equation (S10) captures the dynamics of the algae population $F_u(t)$, which serves as a shared food resource for both native and invasive species. The consumption terms

$\sum_{\Bbbk\in\{\mathbb{n},\mathbb{i}\}} f^{\Bbbk}_{S,u} F_u(t)\left(S^{\Bbbk}_u(t) + \xi_{J,u} J^{\Bbbk}_u(t)\right) dt$ represent the reduction of algae due to feeding by native and invasive susceptible individuals and juveniles, respectively. The parameter $\xi_{J,u}$ denotes the relative ratio of juveniles on algae consumption compared to susceptible adults. The term $\mu\, dt$ represents the refill rate of algae, contributing to its replenishment. The stochastic fluctuation $F_u(t)\, d\zeta_{F,u}$ introduces randomness into the algae dynamics, where $d\zeta_{F,u}$ is the Brownian motion with variance $\sigma^2_{F,u}\, dt$.

The following Figure S15 presents the flow diagram of the SRJF2 model, illustrating the interactions among susceptible individuals ($S^{\mathbb{n}}$, $S^{\mathbb{i}}$), juveniles ($J^{\mathbb{n}}$,$J^{\mathbb{i}}$), algae as a food resource (F), and mortality (R). The bidirectional arrows between $J^{\Bbbk}$ and $S^{\Bbbk}$, represent the maturation of juveniles into susceptible individuals and the reproduction of susceptible individuals into juveniles. Both susceptible individuals and juveniles, consume resources from the alga population F, as indicated by the arrows from F to each population node. The algal food resource is regularly replenished, depicted by the recycling loop, and contributes to the growth of both susceptible individuals and juveniles. All populations have pathways leading to the mortality state R, representing natural death and disease-induced mortality. Two species of *Daphnia* are competing through the food resources in this model.

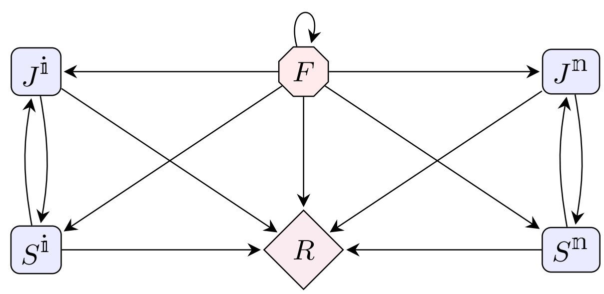


Figure S15: Flow diagram for the SRJF2 model illustrating population interactions. The model includes the R state, representing mortality. For $\Bbbk \in \{\mathbb{n}, \mathbb{i}\}$, Susceptible populations ($S^{\Bbbk}$) can reproduce into juvenile ($J^{\Bbbk}$). And A. Algae (F) will be refilled, as shown by the recycling arrows. Both $S^{\Bbbk}$ and $J^{\Bbbk}$ consume resources from F, and over time, components in F also progress to R.

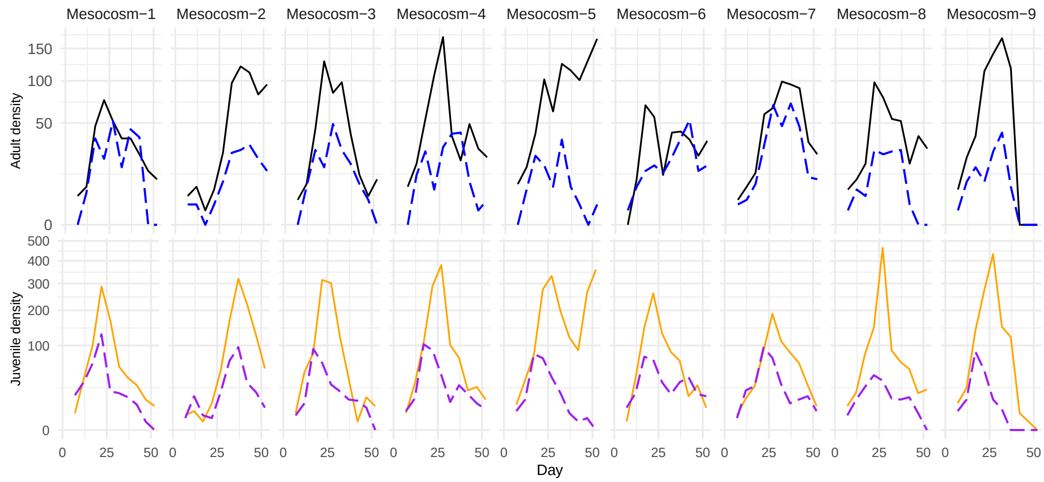


Figure S16: Density (Individuals/Liter) of *D. dentifera* (solid lines) and *D. lumholtzi* (dashed lines). The top panel shows adult susceptibles (*D. dentifera*, black and *D. lumholtzi*, blue). The bottom panel shows juvenile susceptibles (*D. dentifera*, orange; *D. lumholtzi*, purple). There were negligible infected juveniles. Columns are buckets corresponding to replications with same treatment setting.

Parameter	Definition	Unit	Value	CI
$S^{\mathbb{k}}$	Susceptible host density for species $\mathbb{k}$	$\text{individual} \cdot L^{-1}$	Variable	
$J^{\mathbb{k}}$	Juvenile host density for species $\mathbb{k}$	$\text{individual} \cdot L^{-1}$	Variable	
F	Alga density	$10^6 \cdot \text{cell} \cdot L^{-1}$	Variable	
P	Spore density	$10^3 \cdot \text{spore} \cdot L^{-1}$	Variable	
$r^{\mathbb{n}}$	Birth rate of native juvenile	$\text{individual} \cdot 10^{-6} \cdot \text{cell}^{-1}$	$6.29 \cdot 10^3$	$(2.42 \cdot 10^3, 3.76 \cdot 10^4)$
$r^{\mathbb{i}}$	Birth rate of invasive juvenile	$\text{individual} \cdot 10^{-6} \cdot \text{cell}^{-1}$	$4.75 \cdot 10^3$	$(1.78 \cdot 10^3, 6.79 \cdot 10^4)$
$f_S^{\mathbb{n}}$	Native susceptible adult host filtering rate	$\text{L} \cdot \text{individual}^{-1} \cdot \text{day}^{-1}$	$7.10 \cdot 10^{-5}$	$(1.54 \cdot 10^{-5}, 1.17 \cdot 10^{-4})$
$f_S^{\mathbb{i}}$	Invasive susceptible adult host filtering rate	$\text{L} \cdot \text{individual}^{-1} \cdot \text{day}^{-1}$	$8.90 \cdot 10^{-5}$	$(0, 2.04 \cdot 10^{-4})$
$r^{\mathbb{n}} \cdot f_{S^{\mathbb{n}}}$	Effective Birth rate of native juvenile	$10^{-6} \cdot \text{L} \cdot \text{cell}^{-1} \cdot \text{day}^{-1}$	$4.47 \cdot 10^{-1}$	$(3.18 \cdot 10^{-1}, 1.16)$
$r^{\mathbb{i}} \cdot f_{S^{\mathbb{i}}}$	Effective Birth rate of invasive juvenile	$10^{-6} \cdot \text{L} \cdot \text{cell}^{-1} \cdot \text{day}^{-1}$	$4.23 \cdot 10^{-1}$	$(2.67 \cdot 10^{-1}, 3.55)$
$\theta_S^{\mathbb{n}}$	Native susceptible adult host mortality rate	day^{-1}	1.03	$(4.63 \cdot 10^{-1}, 1.86)$
$\theta_S^{\mathbb{i}}$	Invasive susceptible adult host mortality rate	day^{-1}	$4.49 \cdot 10^{-1}$	$(3.12 \cdot 10^{-1}, 1.40)$
$\theta_J^{\mathbb{n}}$	Native juvenile mortality rate	day^{-1}	$2.13 \cdot 10^{-1}$	$(7.94 \cdot 10^{-2}, 9.35 \cdot 10^{-1})$
$\theta_J^{\mathbb{i}}$	Invasive juvenile mortality rate	day^{-1}	$7.35 \cdot 10^{-1}$	$(2.91 \cdot 10^{-1}, 4.08)$
$\lambda_J^{\mathbb{n}}$	Maturation rate of native juvenile	day^{-1}	$1.00 \cdot 10^{-1}$	
$\lambda_J^{\mathbb{i}}$	Maturation rate of invasive juvenile	day^{-1}	$1.00 \cdot 10^{-1}$	
ξ_J	Ratio of juvenile individual filtering rate	Unitless	1.00	
$\beta^{\mathbb{n}}$	Spores produced per infected native individual	$10^3 \cdot \text{spore} \cdot \text{individual}^{-1} \cdot \text{day}^{-1}$	$3.00 \cdot 10$	
$\beta^{\mathbb{i}}$	Spores produced per infected invasive individual	$10^3 \cdot \text{spore} \cdot \text{individual}^{-1} \cdot \text{day}^{-1}$	$3.00 \cdot 10$	
μ	Alga refilling rate	$10^6 \cdot \text{cell} \cdot \text{L}^{-1} \cdot \text{day}^{-1}$	$3.70 \cdot 10^{-1}$	
δ	Sampling rate	day^{-1}	$1.30 \cdot 10^{-2}$	
$\sigma_S^{\mathbb{n}}$	Standard deviation of Brownian motion of native susceptible adult	$\sqrt{\text{individual} \cdot \text{day}^{-1}}$	0	
$\sigma_S^{\mathbb{i}}$	Standard deviation of Brownian motion of invasive susceptible adult	$\sqrt{\text{individual} \cdot \text{day}^{-1}}$	0	
$\sigma_J^{\mathbb{n}}$	Standard deviation of Brownian motion of native juvenile	$\sqrt{\text{individual} \cdot \text{day}^{-1}}$	$2.79 \cdot 10^{-1}$	$(1.38 \cdot 10^{-1}, 4.45 \cdot 10^{-1})$
$\sigma_J^{\mathbb{i}}$	Standard deviation of Brownian motion of invasive juvenile	$\sqrt{\text{individual} \cdot \text{day}^{-1}}$	$4.99 \cdot 10^{-4}$	$(5.87 \cdot 10^{-5}, 7.10 \cdot 10^{-2})$
σ_F	Standard deviation of Brownian motion of alga	$\sqrt{\text{individual} \cdot \text{day}^{-1}}$	$5.37 \cdot 10^{-2}$	$(2.72 \cdot 10^{-2}, 9.19 \cdot 10^{-2})$
$\tau_S^{\mathbb{n}}$	Measurement dispersion for susceptible native adult	Unitless	$1.12 \cdot 10$	$(4.26, 1.10 \cdot 10^2)$
$\tau_S^{\mathbb{i}}$	Measurement dispersion for susceptible invasive adult	Unitless	2.43	$(1.55, 4.44)$

TABLE S8

Variables and parameter definitions and estimates of SRJF2 model for dynamics with both species of Daphnia. The confidence interval is generated by the Monte Carlo Adjusted Profile.

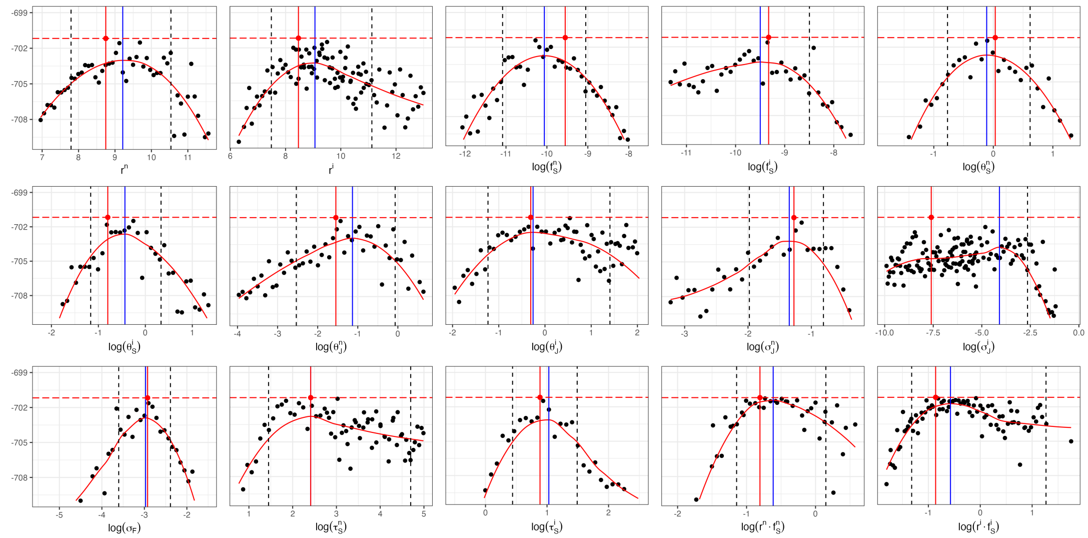


Figure S17: Monte Carlo Adjusted Profile results of SRJF2 model for dynamics with both species of *Daphnia*. The vertical dotted lines represent the 95% confidence interval obtained by MCAP. The vertical blue lines show the MLE estimated using MCAP. The red vertical lines correspond to the model with the overall highest likelihood among all searches.

Specific parameters	Max loglik (PIF)	AIC (PIF)	Max loglik (MPIF)	AIC (MPIF)	LRT p-value (χ^2)	LRT p-value (bootstrap)
$\varnothing$	-701.17	1428.34	-701.17	1428.34	—	—
$\theta^{\mathbb{k}}_{S,u}$	-705.50	1468.99	-691.20	1440.39	0.22	0.81
$f^{\mathbb{k}}_{S,u}$	-720.72	1499.44	-692.71	1443.43	0.39	0.95
$r^{\mathbb{k}}_{u}$	-725.74	1509.49	-696.35	1450.70	0.88	>0.99

TABLE S9

Comparison of model fit and complexity across various configurations of unit-specific parameters within the panelPOMP framework with SRJF2 model for dynamics with both species of Daphnia. The unit-specific parameter setting assessed include (1)$\theta^{\mathbb{k}}_{S,u}$, (2)$f^{\mathbb{k}}_{S,u}$, and (3)$r^{\mathbb{k}}_{u}$ for $\mathbb{k} \in \{\mathbb{n}, \mathbb{i}\}$.

S8.1. *Results.* Table S9 presents a comparison of model fit and complexity across various configurations of unit-specific parameters within the panelPOMP framework for the SRJF2 model, which captures the dynamics of both native and invasive *Daphnia* species. We employed MCAP methods to estimate the maximum log-likelihood and AIC for each model configuration. The models assessed include those with unit-specific parameters for the mortality rates ($\theta^{\mathbb{k}}_{S,u}$), filter rates ($f^{\mathbb{k}}_{S,u}$), and growth efficiency factors ($r^{\mathbb{k}}_{u}$) for $\mathbb{k} \in \{\mathbb{n}, \mathbb{i}\}$. The model with all parameters shared (no unit-specific parameters) achieves the minimum AIC, indicating the best balance between model fit and complexity. The LRT p-values confirm that none of the unit-specific parameterizations provide a statistically significant improvement over the all-shared baseline, corroborating the AIC-based selection. Consequently, we choose to retain the model with all shared parameters, as it offers a more parsimonious representation of the system dynamics without compromising the explanatory power.

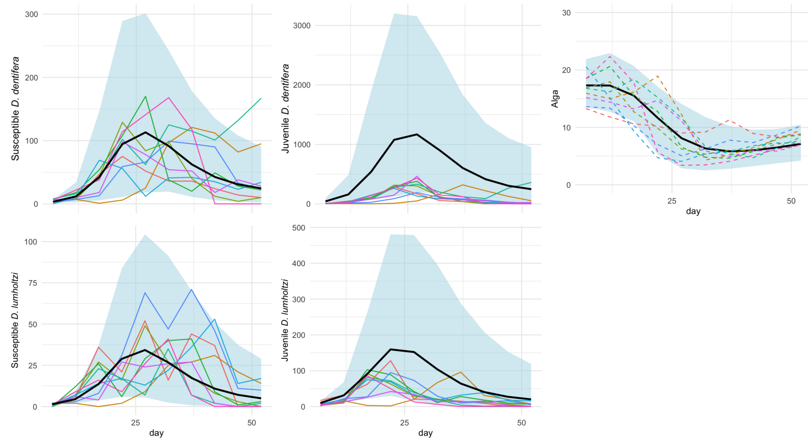


Figure S18: Simulation plots of densities (Individuals/Liter) for susceptible and juvenile populations of both native *D. dentifera* and invasive *D. lumholtzi*, along with alga density (10^6 cells/Liter) over time (days). Colorful solid lines represent observed experimental data for each experimental replicate, showcasing variability across replicates. The black line represents the mean simulation trajectory, while the light blue shaded area corresponds to the 95% confidence interval of the simulation. Dashed lines indicate individual simulation cases, illustrating the model's capacity to capture fluctuations around the mean trajectory.

S8.2. *Simulation.* The simulation results presented in Figures S18 to S19 showcase our model's ability to represent the complex population interactions between *D. dentifera* and *D. lumholtzi*. The model captures the patterns of susceptible and juvenile densities across multiple experimental units, as evidenced by the close alignment of the simulated mean trajectories (illustrated by the black line and light blue confidence bands) with the experimental data (shown by the colorful solid lines). Despite the fact that juvenile density data were not used in calibrating the model, the model accurately predicts juvenile density trends, indicating its capability to generalize ecological processes effectively. The 95% confidence intervals largely encompass the observed juvenile data, supporting the construction of the model's latent process.

Figure S19 provides further insight into the dynamics of algal depletion over time, an aspect that is integral to the model's predictions for *Daphnia* populations. The observed decrease in algal density at later stages, captured in the simulations, suggests a shift to a resource-limited environment—a behavior that aligns well with our model's assumptions. As algae availability declines, population growth of both susceptible and juvenile *Daphnia* becomes constrained, providing a mechanistic explanation for the observed patterns in their densities. By incorporating resource-consumption feedback into the population dynamics, the model demonstrates not only an ability to simulate individual population trends but also to capture this ecological feedback loop.

S9. SRJF: A Model Without Parasites. In this section, we introduce the SRJF model, which formalizes the dynamics among susceptible individuals ($S^{\mathbb{k}}$), the juvenile population ($J^{\mathbb{k}}$), and algae serving as a food resource (F) as follows:

$$dS^{\mathbb{k}}_u(t) = \lambda^{\mathbb{k}}_{J,u} J^{\mathbb{k}}_u(t)\,dt - (\theta^{\mathbb{k}}_{S,u} + \delta) S^{\mathbb{k}}_u(t)\,dt + S^{\mathbb{k}}_u(t)\,d\zeta^{\mathbb{k}}_{S,u}, \tag{S12}$$

$$dJ^{\mathbb{k}}_u(t) = r^{\mathbb{k}}_u f^{\mathbb{k}}_{S,u} F_u(t) S^{\mathbb{k}}_u(t)\,dt - \theta^{\mathbb{k}}_{J,u} J^{\mathbb{k}}_u(t)\,dt - \delta J^{\mathbb{k}}_u(t)\,dt - \lambda^{\mathbb{k}}_{J,u} J^{\mathbb{k}}_u(t)\,dt + J^{\mathbb{k}}_u(t)\,d\zeta^{\mathbb{k}}_{J,u}, \tag{S13}$$

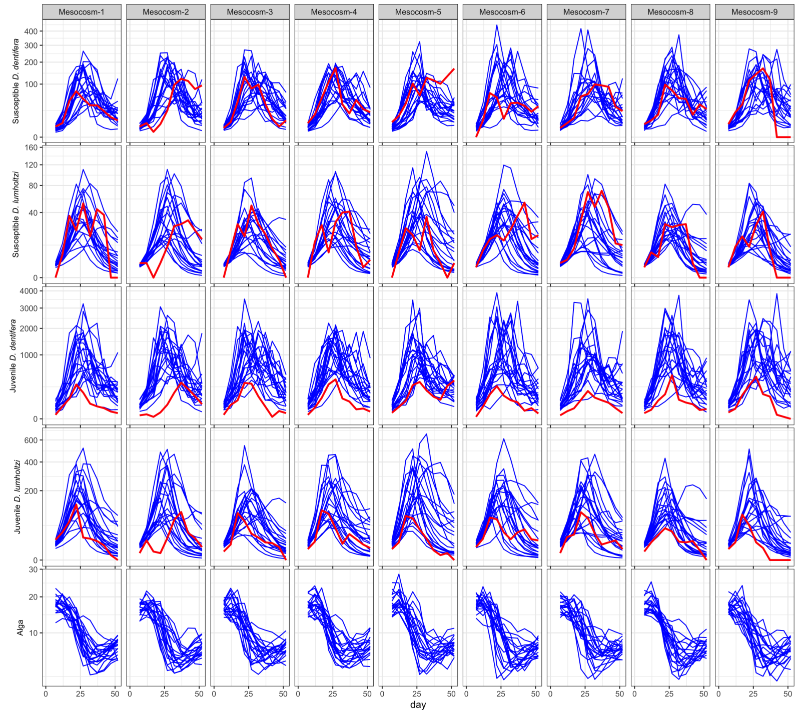


Figure S19: Simulated density (10^6 cells/Liter) of susceptible native *D. dentifera*, susceptible invasive *D. lumholtzi*, juvenile native *D. dentifera*, invasive *D. lumholtzi*, and alga density (F) over time (days) for each experimental unit. The blue lines represent individual simulation cases, while the red line denotes observed experimental data, showing how simulated variability compares to actual data.

$$\text{(S14)} \quad dF_u(t) = -f^{\mathbb{k}}_{S,u} F_u(t) \left(S^{\mathbb{k}}_u(t) + \xi_{J,u} J^{\mathbb{k}}_u(t) \right) dt + \mu \, dt + F_u(t) \, d\zeta_{F,u},$$

$$\text{(S15)} \quad d\zeta^{\mathbb{k}}_{S,u} \sim \mathrm{N}\big[0, (\sigma^{\mathbb{k}}_{S,u})^2 \, dt\big] \; d\zeta^{\mathbb{k}}_{J,u} \sim \mathrm{N}\big[0, (\sigma^{\mathbb{k}}_{J,u})^2 \, dt\big] \; d\zeta_{F,u} \sim \mathrm{N}\big[0, \sigma^2_{F,u} \, dt\big].$$

Equation (S12) describes the rate of change of the susceptible population $S^{\mathbb{k}}_u(t)$. The term $\lambda^{\mathbb{k}}_{J,u} J^{\mathbb{k}}(t) \, dt$ represents the growth of susceptible individuals from the maturation of juvenile population at a rate $\lambda^{\mathbb{k}}_{J,u}$. The loss term $-\left(\theta^{\mathbb{k}}_{S,u} + \delta\right) S^{\mathbb{k}}_u(t) \, dt$ accounts for the mortality of susceptible individuals due to natural causes at rate $\theta^{\mathbb{k}}_{S,u}$ and sampling at rate δ. The stochastic component $S^{\mathbb{k}}_u(t) \, d\zeta^{\mathbb{k}}_{S,u}$ introduces random fluctuations into the susceptible population dynamics, with $d\zeta^{\mathbb{k}}_{S,u}$ being a Brownian motion characterized by variance $(\sigma^{\mathbb{k}}_{S,u})^2 \, dt$. For consistency with the SIRJPF2 and SIRJPF model, we further fix $\sigma^{\mathbb{k}}_{S,u}$ to be zero for specific analysis of this dynamics.

In Equation (S13), the dynamics of the juvenile population $J^{\mathbb{k}}_u(t)$ are modeled. The growth term $r^{\mathbb{k}}_u f^{\mathbb{k}}_{S,u} F^{\mathbb{k}}_u(t) S^{\mathbb{k}}_u(t) \, dt$ signifies the increase in juveniles resulting from feeding on alga, where $r^{\mathbb{k}}_u$ is the growth efficiency factor and $f^{\mathbb{k}}_{S,u}$ is the filtering rate of susceptible individuals. The interaction $F^{\mathbb{k}}_u(t) S^{\mathbb{k}}_u(t)$ captures how the availability of algae influences juvenile growth. The loss term $-\left(\theta^{\mathbb{k}}_{J,u} + \delta + \lambda^{\mathbb{k}}_{J,u}\right) J^{\mathbb{k}}_u(t) \, dt$ reflects natural juvenile mortality at rate $\theta^{\mathbb{k}}_{J,u}$, sampling at rate δ, and the maturing of juveniles to susceptible individuals at rate $\lambda^{\mathbb{k}}_{J,u}$. The stochastic term $J^{\mathbb{k}}_u(t) \, d\zeta^{\mathbb{k}}_{J,u}$ reflects random environmental and demographic fluctuations affecting juveniles, with $d\zeta^{\mathbb{k}}_{J,u}$ being Brownian motion with variance $(\sigma^{\mathbb{k}}_{J,u})^2\,dt$. However, introducing several stochastic terms when describing *Daphnia* behavior allows too much freedom for model, which results in weakly identified parameters and overfitting. For the purpose of better generalization, we allow the stochastic term in juvenile states to affect the density of adult to show the fluctuations.

Equation (S14) captures the dynamics of the *A.* algae population $F^{\mathbb{k}}_u(t)$. The consumption term $-f^{\mathbb{k}}_{S,u}F^{\mathbb{k}}_u(t)\left(S^{\mathbb{k}}_u(t)+\xi_{J,u}J^{\mathbb{k}}_u(t)\right)dt$ represents the reduction of algae due to feeding by both susceptible individuals and juveniles, with $\xi_{J,u}$ indicating the ratio of juveniles on algae consumption compared to susceptible individuals. The term $\mu\,dt$ denotes the refill rate of alga. The stochastic fluctuation $F^{\mathbb{k}}_u(t)\,d\zeta^{\mathbb{k}}_{F,u}$ introduces randomness into the algae dynamics, where $d\zeta^{\mathbb{k}}_{F,u}$ shows the Brownian motion with variance $\sigma^2_{F,u}\,dt$. These terms model the latent randomness in the system due to environmental variability and other stochastic factors influencing the susceptible individuals, juveniles, and alga populations.

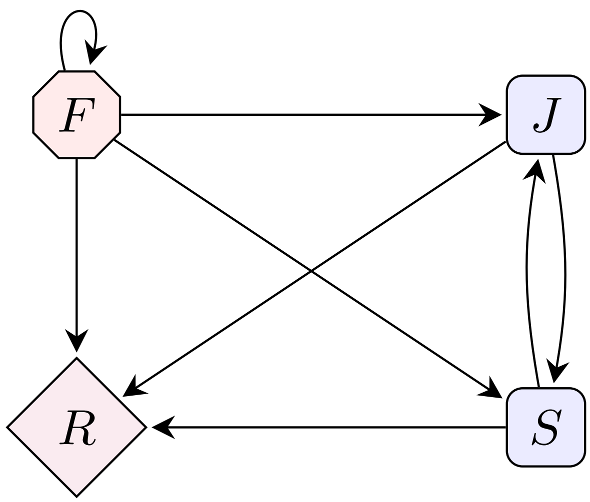


Figure S20: Flow diagram for the SRJF model illustrating population interactions. The model includes the R state, representing mortality. For $\mathbb{k}\in\{\mathtt{m},\mathtt{i}\}$, susceptible populations ($S^{\mathbb{k}}$) can reproduce into juvenile ($J^{\mathbb{k}}$) and *A.* Algae (F) will be refilled, as shown by the recycling arrows. Both $S^{\mathbb{k}}$ and $J^{\mathbb{k}}$ consume resources from F, and over time, components in F also progress to R.

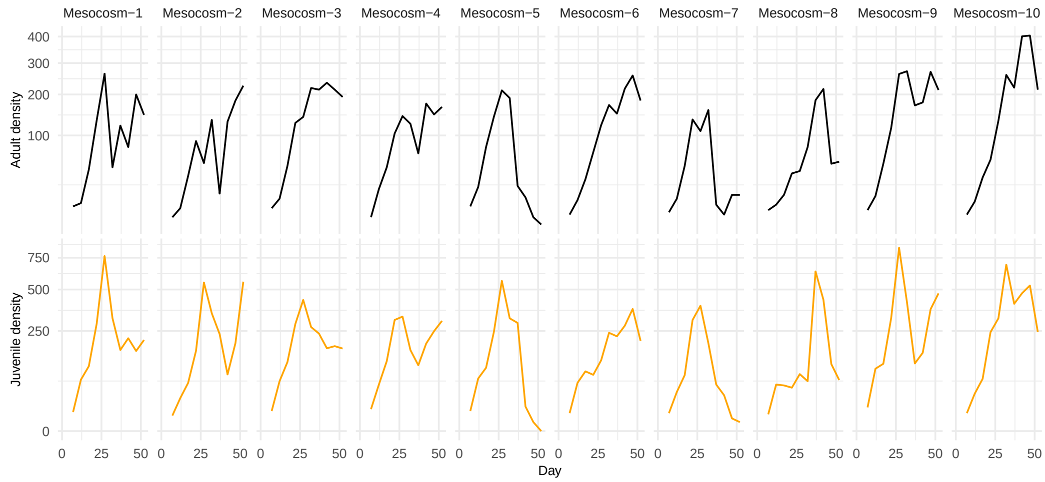


Figure S21: Density (Individuals/Liter) of *D. dentifera*. The top panel shows adult susceptibles (*D. dentifera*, black). The bottom panel shows juvenile susceptibles (*D. dentifera*, orange). There were negligible infected juveniles. Columns are buckets corresponding to replications with same treatment setting.

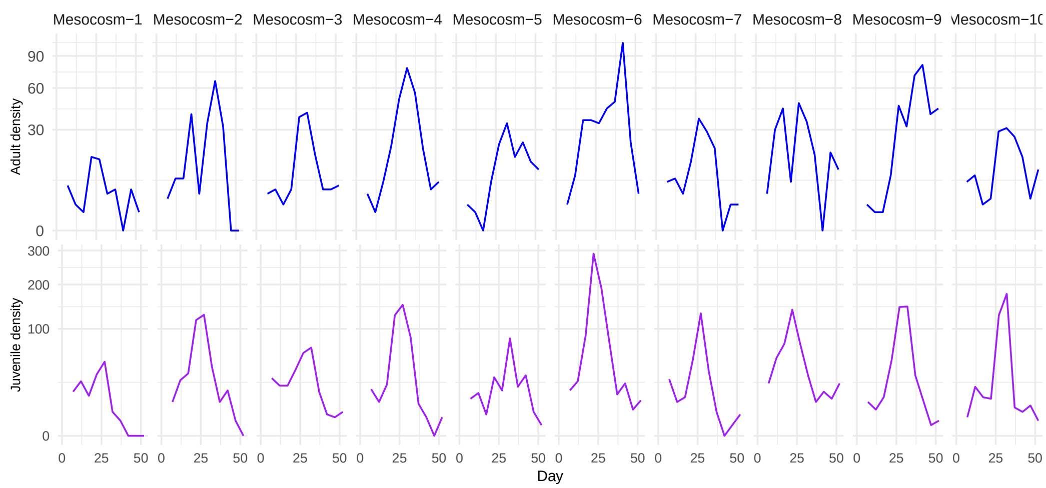


Figure S22: Density (Individuals/Liter) of *D. lumholtzi*. The top panel shows adult susceptibles (*D. lumholtzi*, blue). The bottom panel shows juvenile susceptibles (*D. lumholtzi*, purple). There were negligible infected juveniles. Columns are buckets corresponding to replications with same treatment setting.

Parameter	Definition	Unit	Value	CI
$S^{\mathbb{n}}$	Susceptible host density	$\text{individual} \cdot L^{-1}$	Variable	
$J^{\mathbb{n}}$	Juvenile host density	$\text{individual} \cdot L^{-1}$	Variable	
F	Alga density	$10^6 \cdot \text{cell} \cdot L^{-1}$	Variable	
$r^{\mathbb{n}}$	Birth rate of juvenile	$\text{individual} \cdot 10^{-6} \cdot \text{cell}^{-1}$	$1.69 \cdot 10^3$	$(6.78 \cdot 10^2, 4.53 \cdot 10^3)$
$f_S^{\mathbb{n}}$	Susceptible adult host filtering rate	$\text{L} \cdot \text{individual}^{-1} \cdot \text{day}^{-1}$	$1.32 \cdot 10^{-4}$	$(7.29 \cdot 10^{-5}, 2.39 \cdot 10^{-4})$
$r^{\mathbb{n}} \cdot f_{S^{\mathbb{n}}}$	Effective Birth rate of native juvenile	$10^{-6} \cdot \text{L} \cdot \text{cell}^{-1} \cdot \text{day}^{-1}$	$2.23 \cdot 10^{-1}$	$(1.53 \cdot 10^{-1}, 3.59 \cdot 10^{-1})$
$\theta_S^{\mathbb{n}}$	Susceptible adult host mortality rate	day^{-1}	$6.92 \cdot 10^{-1}$	$(4.27 \cdot 10^{-1}, 1.27)$
$\theta_J^{\mathbb{n}}$	Juvenile mortality rate	day^{-1}	$2.13 \cdot 10^{-8}$	$(0, 4.56 \cdot 10^{-2})$
$\lambda_J^{\mathbb{n}}$	Maturation rate of the juvenile	day^{-1}	$1.00 \cdot 10^{-1}$	
ξ_J	Ratio of juvenile individual filtering rate	Unitless	1.00	
$\beta^{\mathbb{n}}$	Spores produced by death per infected individual	$10^3 \cdot \text{spore} \cdot \text{individual}^{-1} \cdot \text{day}^{-1}$	$3.00 \cdot 10$	
μ	Algae refilling rate	$10^6 \cdot \text{cell} \cdot \text{L}^{-1} \cdot \text{day}^{-1}$	$3.70 \cdot 10^{-1}$	
δ	Sampling rate	day^{-1}	$1.30 \cdot 10^{-2}$	
$\sigma_S^{\mathbb{n}}$	Standard deviation of Brownian motion of susceptible adult	$\sqrt{\text{individual} \cdot \text{day}^{-1}}$	0	
$\sigma_J^{\mathbb{n}}$	Standard deviation of Brownian motion of juvenile	$\sqrt{\text{individual} \cdot \text{day}^{-1}}$	$3.07 \cdot 10^{-1}$	$(2.31 \cdot 10^{-1}, 3.81 \cdot 10^{-1})$
σ_F	Standard deviation of Brownian motion of alga	$\sqrt{\text{individual} \cdot \text{day}^{-1}}$	$9.31 \cdot 10^{-7}$	$(0, 4.60 \cdot 10^{-2})$
$\tau_S^{\mathbb{n}}$	Measurement dispersion for susceptible adult	Unitless	$1.34 \cdot 10$	$(6.77, 9.81 \cdot 10)$

TABLE S10

Variables and parameter definitions and estimates of SRJF model for D. dentifera-only dynamics. The confidence interval is generated by the Monte Carlo Adjusted Profile.

Parameter	Definition	Unit	Value	CI
$S^{\mathbb{i}}$	Susceptible host density	$\text{individual} \cdot L^{-1}$	Variable	
$J^{\mathbb{i}}$	Juvenile host density	$\text{individual} \cdot L^{-1}$	Variable	
F	Alga density	$10^6 \cdot \text{cell} \cdot L^{-1}$	Variable	
$r^{\mathbb{i}}$	Birth rate of juvenile	$\text{individual} \cdot 10^{-6} \cdot \text{cell}^{-1}$	$9.77 \cdot 10^2$	$(8.86 \cdot 10, \infty)$
$f_S^{\mathbb{i}}$	Susceptible adult host filtering rate	$\text{L} \cdot \text{individual}^{-1} \cdot \text{day}^{-1}$	$2.67 \cdot 10^{-4}$	$(0, 1.34 \cdot 10^{-3})$
$r^{\mathbb{i}} \cdot f_{S^{\mathbb{i}}}$	Effective Birth rate of invasive juvenile	$10^{-6} \cdot \text{L} \cdot \text{cell}^{-1} \cdot \text{day}^{-1}$	$2.61 \cdot 10^{-1}$	$(9.04 \cdot 10^{-2}, 4.02)$
$\theta_S^{\mathbb{i}}$	Susceptible adult host mortality rate	day^{-1}	$5.99 \cdot 10^{-1}$	$(2.26 \cdot 10^{-1}, 1.40)$
$\theta_J^{\mathbb{i}}$	Juvenile mortality rate	day^{-1}	$2.98 \cdot 10^{-1}$	$(0, 4.58)$
$\lambda_J^{\mathbb{i}}$	Maturation rate of the juvenile	day^{-1}	$1.00 \cdot 10^{-1}$	
ξ_J	Ratio of juvenile individual filtering rate	Unitless	1.00	
$\beta^{\mathbb{i}}$	Spores produced by death per infected individual	$10^3 \cdot \text{spore} \cdot \text{individual}^{-1} \cdot \text{day}^{-1}$	$3.00 \cdot 10$	
μ	Algae refilling rate	$10^6 \cdot \text{cell} \cdot \text{L}^{-1} \cdot \text{day}^{-1}$	$3.70 \cdot 10^{-1}$	
δ	Sampling rate	day^{-1}	$1.30 \cdot 10^{-2}$	
$\sigma_S^{\mathbb{i}}$	Standard deviation of Brownian motion of susceptible adult	$\sqrt{\text{individual} \cdot \text{day}^{-1}}$	0	
$\sigma_J^{\mathbb{i}}$	Standard deviation of Brownian motion of juvenile	$\sqrt{\text{individual} \cdot \text{day}^{-1}}$	$3.78 \cdot 10^{-1}$	$(0, 5.07 \cdot 10^{-1})$
$\sigma_F^{\mathbb{i}}$	Standard deviation of Brownian motion of alga	$\sqrt{\text{individual} \cdot \text{day}^{-1}}$	$4.73 \cdot 10^{-2}$	$(0, 1.38 \cdot 10^{-1})$
$\tau_S^{\mathbb{i}}$	Measurement dispersion for susceptible adult	Unitless	2.35	$(1.32, 5.72)$

TABLE S11

Variables and parameter definitions and estimates of SRJF model for D. lumholtzi-only dynamics. The confidence interval is generated by the Monte Carlo Adjusted Profile.

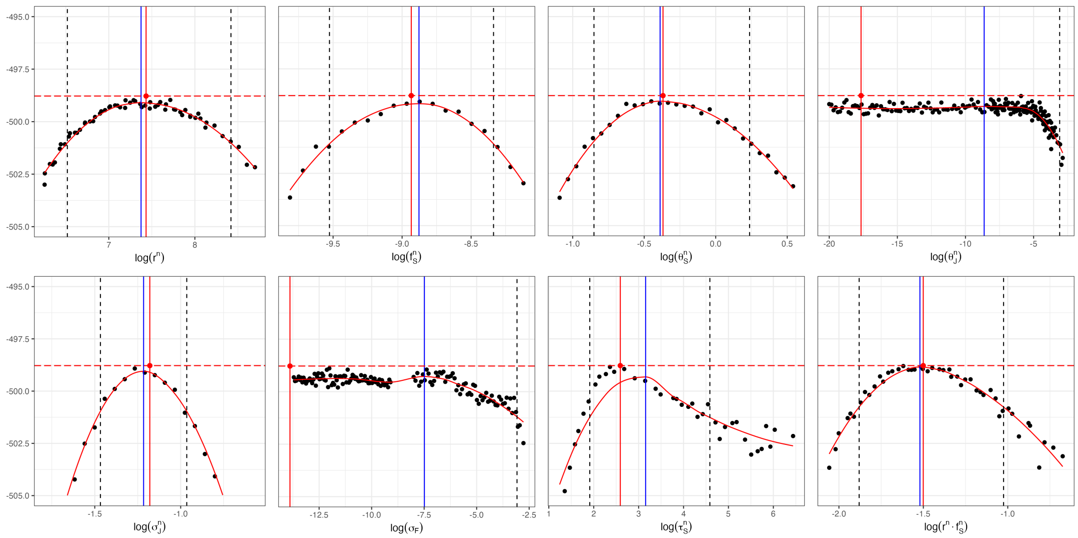


Figure S23: Monte Carlo Adjusted Profile results of SRJF model for *D. dentifera*-only dynamics. The vertical dotted lines represent the 95% confidence interval obtained by MCAP. The vertical blue lines show the MLE estimated using MCAP. The red vertical lines correspond to the model with the overall highest likelihood among all searches.

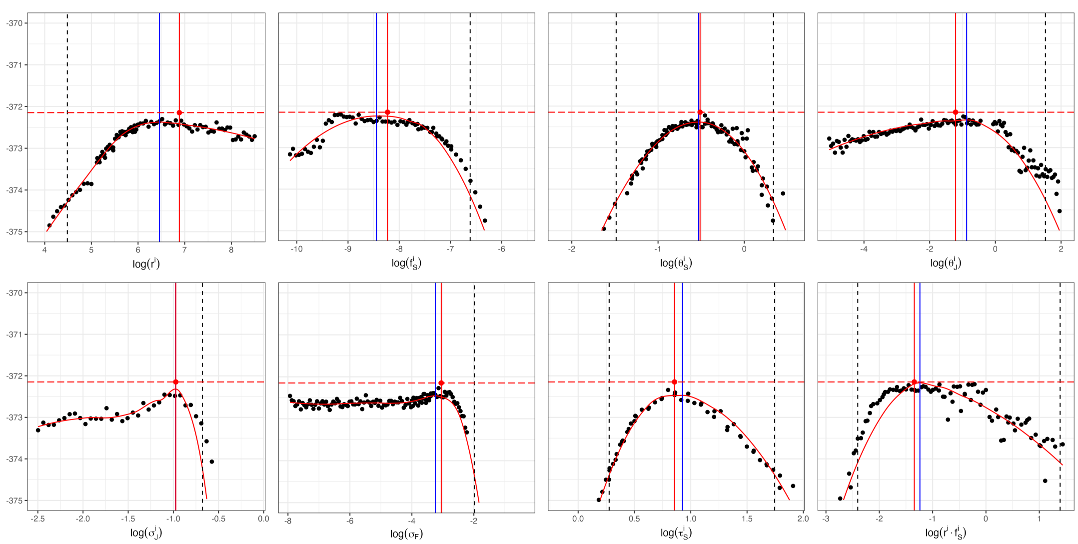


Figure S24: Monte Carlo Adjusted Profile results of SRJF model for *D. lumholtzi*-only dynamics. The vertical dotted lines represent the 95% confidence interval obtained by MCAP. The vertical blue lines show the MLE estimated using MCAP. The red vertical lines correspond to the model with the overall highest likelihood among all searches.

S9.1. *Results.* The data plots, parameter estimates, and profile likelihood plots are analogous to the results in the main text, applied here to the SRJF model for either *D. dentifera* or *D. lumholtzi*.

Specific parameters	Max loglik (PIF)	AIC (PIF)	Max loglik (MPIF)	AIC (MPIF)	LRT p-value (χ^2)	LRT p-value (bootstrap)
$\varnothing$	-498.78	1011.56	-498.78	1011.56	—	—
$\theta_{S,u}^{\mathbb{m}}$	-490.57	1013.14	-489.85	1011.71	0.04	0.12
$r_u^{\mathbb{m}}$	-497.43	1026.85	-496.44	1024.89	0.86	0.94
$f_{S,u}^{\mathbb{m}}$	-497.43	1026.86	-496.16	1024.32	0.81	0.91

TABLE S12

Comparison of model fit and complexity across various configurations of unit-specific parameters within the panelPOMP framework for D. dentifera-only dynamics. The unit-specific parameter setting assessed include $f_{S,u}^{\mathbb{m}}$, $r_u^{\mathbb{m}}$, $\theta_{S,u}^{\mathbb{m}}$.

Specific parameters	Max loglik (PIF)	AIC (PIF)	Max loglik (MPIF)	AIC (MPIF)	LRT p-value (χ^2)	LRT p-value (bootstrap)
$\varnothing$	-372.15	758.294	-372.15	758.294	—	—
$\theta_{S,u}^{\mathbb{l}}$	-365.33	762.658	-365.37	762.733	0.14	0.42
$r_u^{\mathbb{l}}$	-366.84	765.676	-366.29	764.571	0.23	0.65
$f_{S,u}^{\mathbb{l}}$	-368.15	768.300	-368.07	768.147	0.52	0.84

TABLE S13

Comparison of model fit and complexity across various configurations of unit-specific parameters within the panelPOMP framework for D. lumholtzi-only dynamics. The unit-specific parameter setting assessed include $f_{S,u}^{\mathbb{l}}$, $r_u^{\mathbb{l}}$, $\theta_{S,u}^{\mathbb{l}}$.

The results in Tables S12 and S13 indicate that the model configuration with all shared parameters achieves the minimum AIC score for both dynamics, suggesting that this model makes a successful balance between model fit and complexity. The LRT p-values further support this choice, as none of the unit-specific models show a statistically significant improvement over the all-shared baseline. These results are consistent with our other analysis.

S9.2. *Simulation.* The simulation plots (Figures S25 to S28) collectively demonstrate the SRPF model's capability to capture the complex interactions among various states within the population dynamics of *D. dentifera* and *D. lumholtzi*. Specifically, the model effectively reproduces the latent patterns for susceptible and infected densities across multiple experimental units, as evidenced by the close alignment between the simulated mean trajectories (depicted by the black and light-blue bands) and the observed experimental data (represented by colorful solid lines). Each blue line illustrates an individual simulation case, capturing the stochastic variability across replicates, while the light blue shaded area (where present) denotes the 95% confidence interval of the simulation results. Although juvenile density data were not included in the model fitting process, the model still accurately predicts juvenile density trends (Figures S28). This performance indicates the model's capacity to generalize underlying ecological processes, as the 95% confidence intervals largely encompass the observed juvenile data. The model also predicts a decline in algal density as the experiment progresses. This predicted reduction in food availability provides a mechanistic explanation for the observed trends in *Daphnia* densities, as resource scarcity constrains population growth, ultimately affecting the susceptible, infected, and juvenile densities.

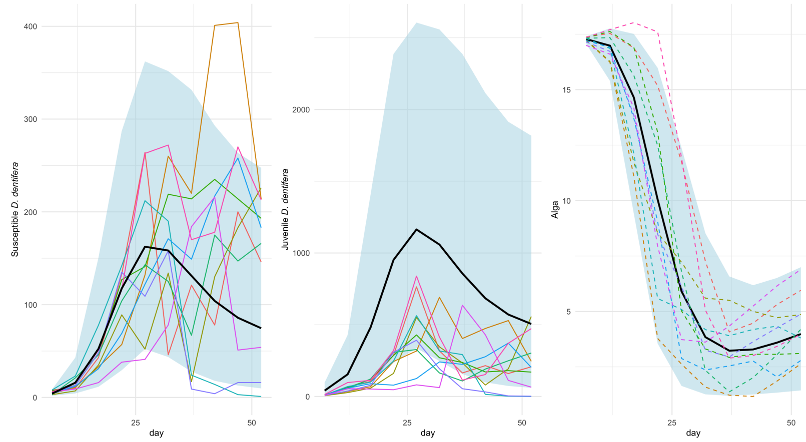


Figure S25: Simulation plots of densities (Individuals/Liter) for susceptible, and juvenile *D. dentifera*, along with alga density (10^6 cells/Liter) and parasite density (10^3 spores/Liter), over time (days). The colorful solid lines represent observed experimental data, showing the variability across different experimental replicates. The black line represents the mean trajectory from the simulation model, capturing the general trend of each density. The light blue shaded area corresponds to the 95% confidence interval of the simulations, indicating the range of model predictions. Dashed lines illustrate individual simulated trajectories, highlighting the model's ability to capture fluctuations around the mean trajectory.

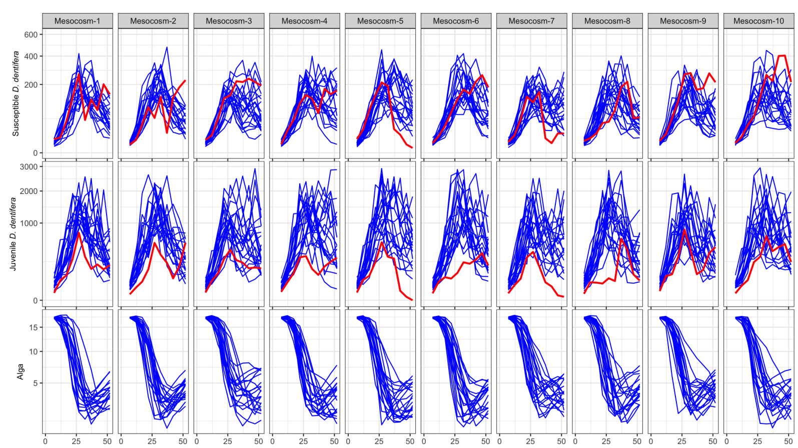


Figure S26: Simulated densities (Individuals/Liter) of susceptible *D. dentifera*, juvenile *D. dentifera* and alga density (10^6 cells/Liter) over time (days) for each experimental unit. The blue lines represent individual simulation runs, capturing the variability in susceptible density across replicates, while the red line represents the actual experiment data.

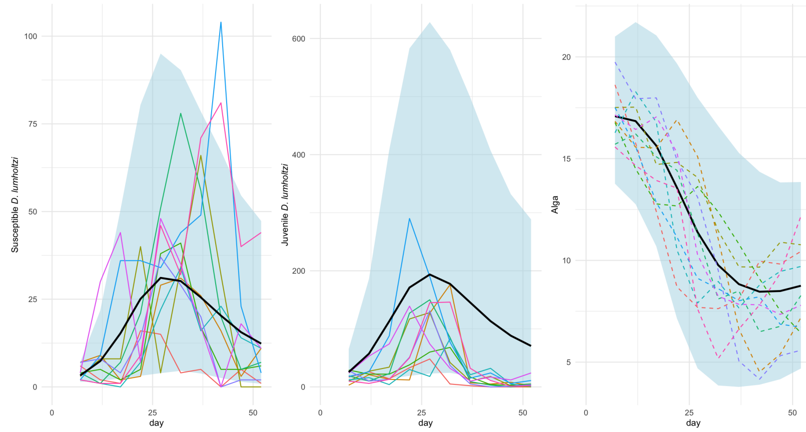


Figure S27: Simulation plots of densities (Individuals/Liter) for susceptible, and juvenile *D. lumholtzi*, along with alga density (10^6·cells/Liter) and parasite density (10^3·spores/Liter), over time (days). The colorful solid lines represent observed experimental data, showing the variability across different experimental replicates. The black line represents the mean trajectory from the simulation model, capturing the general trend of each density. The light blue shaded area corresponds to the 95% confidence interval of the simulations, indicating the range of model predictions. Dashed lines illustrate individual simulated trajectories, highlighting the model's ability to capture fluctuations around the mean trajectory.

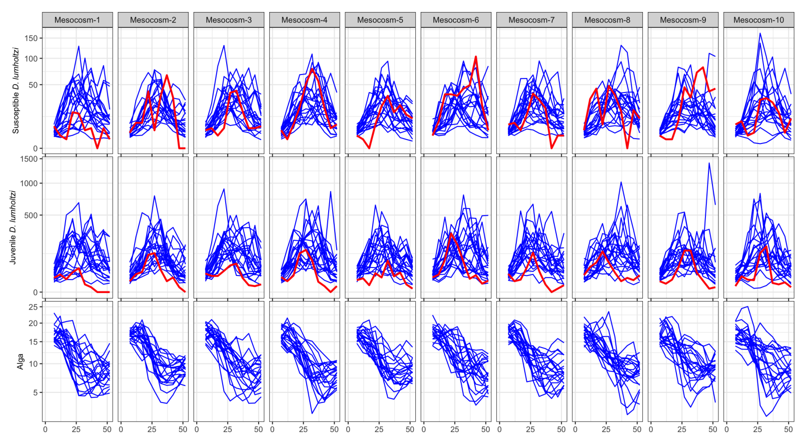


Figure S28: Simulated densities (Individuals/Liter) of susceptible *D. lumholtzi* and juvenile *D. lumholtzi*, with alga density (10^6·cells/Liter), for each experimental unit. The blue lines represent individual simulation runs, capturing the variability in susceptible density across replicates, while the red line represents the actual experiment data.

S10. Benchmark Models.

S10.1. *Searle et al.'s Model.* Using the mechanistic model introduced by Searle et al. (2016), we conduct predictions of population $\tilde{y}_{u,n}$ for susceptible, infected *D. dentifera* and

D. lumholtzi for all time $n \in \{1, 2, \ldots, N_u\}$ and units $u \in \{1, 2, \ldots, U\}$. In our case, each unit is observed at the same number of time points, so $N_u = N$ for all u. Then, we obtain the overall log-likelihood by summing the log-probabilities of the observed density of *Daphnia*. Specifically, we use the observed densities $y^*_{u,n}$ and the predicted means $\tilde{y}_{u,n}$ derived from the mechanistic model, along with the dispersion parameter τ to specify a NBinom distribution and calculate the likelihood:

$$\ell(\beta, \theta) = \sum_{u=1}^{U} \sum_{n=1}^{N} \log P(Y_{u,n} = y^*_{u,n} \mid \mu_{u,n}, \tau), \tag{S16}$$

To obtain the maximum likelihood estimate, we conduct the linear optimization on τ and choose the $\hat{\tau}$ that yields the highest overall likelihood. With this $\hat{\tau}$, we obtain the likelihood by S16 for the model on the data. And the result of the model fitting can be found in the Table 2 in the main article.

S10.2. *Negative Binomial Regression.* Count data from ecological studies often exhibit overdispersion and unit-specific variation. The NBinom distribution addresses overdispersion, while random intercepts allow each unit to have its own baseline level. As a benchmark model, we employ NBinom generalized linear mixed models (GLMMs) to capture overdispersed count data and correlation across repeated measures within each unit u. The NBinom distribution is parameterized by a mean $\mu_{u,n}$ and a dispersion parameter θ.

We present three GLMMs with increasing of order of time factor to capture the trend of these *Daphnia* species in the experiment. While a linear model is easy to interpret and straightforward to implementing, observed *Daphnia* data exhibit a non-linear pattern. Introducing quadratic or cubic terms to the model enhance the capability of model to capture these non-linear trends in the data. By starting with a linear model and then considering quadratic and cubic cases, we can evaluate the complexity of temporal patterns and better approximate the latent process of in the dynamics.

S10.2.1. *Model 1 (Linear Trend Model).* We start with a simple GLMM having only with linear terms:

$$Y_{u,n} \sim \text{NBinom}(\mu_{u,n}, \tau), \tag{S17}$$

$$\log(\mu_{u,n}) = \beta_0 + \beta_1 \, t_{u,n} + b_u, \tag{S18}$$

where $\mu_{u,n}$ is the expected count for unit u at time n. We fit this model independently for each *Daphnia* species, life stage, and infection status, and so the total log-likelihood is the sum over these categories. The vector coefficient β_1 measures the effect of time on the log-scale of *Daphnia* density. The term b_u accounts for the random intercept drawn from a normal distribution with variance σ^2. Conditional on the random intercept b_u, observations within a unit are assumed independent. The dispersion parameter τ from the NBinom distribution accommodates overdispersion in the count data, which is density in our case.

S10.2.2. *Model 2 (Quadratic Trend Model).* To accommodate the non-linear pattern, we include a quadratic term in time in model S20:

$$Y_{u,n} \sim \text{NBinom}(\mu_{u,n}, \tau), \tag{S19}$$

$$\log(\mu_{u,n}) = \beta_0 + \beta_1 \, t_{u,n} + \beta_2 \, t^2_{u,n} + b_u. \tag{S20}$$

Here, β_2 captures curvature in the trend, allowing for a turning point as time progresses.

S10.2.3. *Model 3 (Cubic Trend Model).* To capture more complex temporal pattern, we further introduce the cubic term in the model S22:

$$Y_{u,n} \sim \text{NBinom}(\mu_{u,n}, \tau), \tag{S21}$$

$$\log(\mu_{u,n}) = \beta_0 + \beta_1\, t_{u,n} + \beta_2\, t_{u,n}^2 + \beta_3\, t_{u,n}^3 + b_u. \tag{S22}$$

The cubic term β_3 provides additional capability for capturing multiple fluctuation patterns in the *Daphnia* dynamics.

The likelihood for these models is constructed by considering the joint distribution of all observations $\{Y_{u,n}\}$, integrating over the random effects, $b_{1:U}$. Letting $\beta = (\beta_0, \beta_1, \dots)$ denote the fixed-effect parameters, the conditional likelihood of the observations for unit u given b_u is

$$L_u(\beta, \tau, \sigma^2) = \prod_{n=1}^{N} f_{\text{NB}}\big(Y_{u,n}; \mu_{u,n}(\beta, \sigma^2), \tau\big), \tag{S23}$$

where $f_{\text{NB}}(\cdot; \mu, \tau)$ is the NBinom probability mass function and $\mu_{u,n}(\beta, b_u) = \exp(\beta_0 + \beta_1 t_{u,n} + \cdots + b_u)$. The total likelihood is then the product over all units due to the independence:

$$L(\beta, \tau, \sigma^2) = \prod_{u=1}^{U} L_u(\beta, \tau, \sigma^2) \tag{S24}$$

$$= \prod_{u=1}^{U} \prod_{n=1}^{N} f_{\text{NB}}\big(Y_{u,n}; \mu_{u,n}(\beta, \sigma^2), \tau\big) \tag{S25}$$

and the log-likelihood is:

$$\ell(\beta, \tau, \sigma^2) = \sum_{u=1}^{U} \sum_{n=1}^{N} \log\big(f_{\text{NB}}(Y_{u,n}; \mu_{i,j}(\beta, \sigma^2), \tau)\big). \tag{S26}$$

The calculated log-likelihood used to compute the AIC for model comparison. And the result of the model fitting of model S18, S20 and S22 can be found in the Table 2 in the main article.

S11. SIRJP2: An *F* Ablation of the Full Model. We considered an SIRJP2 model derived by removing the latent food dynamics from the SIRJPF2 model. This ablation experiment tests whether the latent food compartment F is central to the way the model explains the population dynamics. We implemented SIRJP2 by fitting SIRJPF2 while holding F fixed at its initial value $F(t_0) = 16.667$ for the whole trajectory. Effectively, this eliminates from the dynamics food loss due to consumption by *Daphnia*, food renewal inflow, and the process noise σ_F. The birth terms still read the food level through $r^{\Bbbk} f_S^{\Bbbk} F$, but with F held constant the per-capita recruitment rate no longer responds to a depletable resource. We fit the ablated model to the observed data using the same panel iterated-filtering protocol as the all-shared SIRJPF2 model, warm-started from the all-shared maximum likelihood estimate. Dropping σ_F lowers the free-parameter count from 24 to 23, so the ablation is actually favoured by the AIC penalty. Because freezing the food dynamics is a structural change rather than a parameter restriction, the two models are non-nested and we compare them by AIC. Results are shown in Table 2 and copied in Table S14.

Holding the food compartment fixed costs about 7.2 log-likelihood units, even though the ablated model is handed a free parameter back. On the AIC scale the model with a depletable

TABLE S14

Comparison of the all-shared SIRJPF2 model with its F ablation, SIRJP2, in which the food compartment is held fixed at its initial value. Log-likelihoods are evaluated by particle filtering, and the standard error shown for the ablation is the Monte Carlo standard error of the maximized log-likelihood across replicate searches. The no-F model is decisively worse by AIC.

Model	Free parameters k	Max log-likelihood	AIC
All-shared (depletable F)	24	-880.6	1809.1
No-F (F held constant)	23	-887.8 ± 0.16	1821.6
ΔAIC (no-F $-$ all-shared)			$+12.4$

food compartment is preferred by $\Delta\mathrm{AIC} \approx 12.4$, which is decisive on the conventional evidential scale. When the resource does not deplete, recruitment into the juvenile classes $J^{\mathbb{n}}$ and $J^{\mathbb{i}}$ continues at an undiminished rate as the adult population grows, and the model cannot reproduce the resource-limited turnover that drives the observed rise and fall in *Daphnia* abundance. It is exactly the feedback in which a growing population draws down the food supply, which in turn slows its own recruitment, that produces the peak-then-collapse signature in the data. The ablation confirms that explicitly modeling resource depletion through the latent food compartment adds significantly to the model's ability to explain the data.

S12. SIRPF2: A J Ablation of the Full Model. The SIRPF2 model presented here ignores the state of juvenile *Daphnia* and describes the dynamics in each bucket (u) among susceptible individuals ($S^{\mathbb{k}}$), infected individuals ($I^{\mathbb{k}}$), alga as a food resource (F), the parasite population (P). The superscript $\mathbb{k}$ denotes the species, which represents native or invasive species. Each stochastic differential equation accounts for various biological and ecological processes, augmented with stochastic terms to capture the inherent randomness within the systems.

$$dS^{\mathbb{k}}_u(t) = r^{\mathbb{k}}_u\, f^{\mathbb{k}}_{S,u}\, F_u(t)\, S^{\mathbb{k}}_u(t)\, dt - \left\{\theta^{\mathbb{k}}_{S,u} + p^{\mathbb{k}}_u\, f^{\mathbb{k}}_{S,u}\, P_u(t) + \delta\right\} S^{\mathbb{k}}_u(t)\, dt, \tag{S27}$$

$$dI^{\mathbb{k}}_u(t) = p^{\mathbb{k}}_u\, f^{\mathbb{k}}_{S,u}\, S^{\mathbb{k}}_u(t)\, P_u(t)\, dt - \left\{\theta^{\mathbb{k}}_{I,u} + \delta\right\} I^{\mathbb{k}}_u(t)\, dt + I^{\mathbb{k}}_u(t)\, d\zeta^{\mathbb{k}}_{I,u}, \tag{S28}$$

$$dP_u(t) = \sum_{\mathbb{k}\in\{\mathbb{n},\mathbb{i}\}} \left(\beta^{\mathbb{k}}_u\, \theta^{\mathbb{k}}_{I,u}\, I^{\mathbb{k}}_u(t) - f^{\mathbb{k}}_{S,u}\left\{S^{\mathbb{k}}_u(t) + \xi_u I^{\mathbb{k}}_u(t)\right\} P_u(t)\right) dt \\ - \theta_{P,u}\, P_u(t)\, dt + P_u(t)\, d\zeta_{P,u}, \tag{S29}$$

$$dF_u(t) = -\sum_{\mathbb{k}\in\{\mathbb{n},\mathbb{i}\}} f^{\mathbb{k}}_{S,u}\, F_u(t) \left(S^{\mathbb{k}}_u(t) + \xi_u\, I^{\mathbb{k}}_u(t)\right) dt + \mu\, dt + F_u(t)\, d\zeta_{F,u}, \tag{S30}$$

$$d\zeta^{\mathbb{k}}_{I,u} \sim \mathrm{N}\big[0, (\sigma^{\mathbb{k}}_{I,u})^2\, dt\big], \qquad d\zeta_{F,u} \sim \mathrm{N}\big[0, \sigma^2_{F,u}\, dt\big], \qquad d\zeta_{P,u} \sim \mathrm{N}\big[0, \sigma^2_{P,u}\, dt\big]. \tag{S31}$$

Unlike the SIRJPF2 model, we ignore the maturation process from juvenile to adult in equation (S28). Instead, this model uses $r^{\mathbb{k}}_u$ to represent the average rate of growth in density of *Daphnia* adult.

The SIRPF2 model illustrates the interdependent dynamics among the Susceptible (S), Infected (I), Food (F), and Parasite (P) populations, with transitions representing ecological and biological interactions. The model is represented diagrammatically in Fig. S29. The maximized log-likelihood for this model is presented in Table 2.

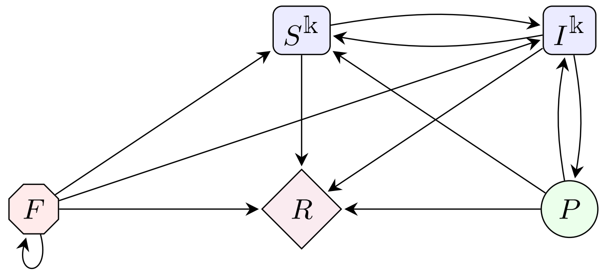


Figure S29: Flow diagram for the SIRPF2 model illustrating population interactions.

Mesocosm	*D. dentifera* juvenile	*D. dentifera* female adult	*D. lumholtzi* juvenile	*D. lumholtzi* female adult	*D. lumholtzi* male adult
1	22	22	22	27	27
2	22	32	17	17	22
3	27	32	27	22	37
4	22	22	37	17	27
5	22	22	17	32	27
6	32	27	27	32	42
7	27	22	17	17	27
8	27	32	22	22	42

TABLE S15

This table shows the day on which each species reached its peak density in mesocosms containing both Daphnia species and the parasite. D. dentifera male adults are excluded because they were almost absent.

S13. Experimental Data on Peak Density. Table S15 provides supporting data on the relative timing of peak abundance for different demographic groups.

S14. SIRJPF2-Gamma Model. The SIRJPF2-Gamma model differs from the original SIRJPF2 model by its incorporation of gamma noise in transition processes between states, in place of the Gaussian noise used previously. Specifically, in the SIRJPF2-Gamma model, random fluctuations for transitions between states are explained by gamma distributions to represent ecological events occur through one-way transitions which are nonnegative. The use of gamma noise in modeling transitions explicitly enforces that individuals move from one class to another without the artificial creation or destruction of hosts that do not transit to or from another compartment. This aligns with the underlying dynamics that the jump-like transitions observed in ecosystem reflect one-directional movements rather than symmetrical fluctuations.

$$dS_u^{\Bbbk}(t) = \lambda_{J,u}^{\Bbbk} J_u^{\Bbbk}(t)dt - \left\{\theta_{S,u}^{\Bbbk} + \delta\right\} S_u^{\Bbbk}(t)\,dt - p_u^{\Bbbk} f_{S,u}^{\Bbbk} P_u(t) S_u^{\Bbbk}(t)\,d\Gamma_{SI,u}^{\Bbbk}, \tag{S32}$$

$$dI_u^{\Bbbk}(t) = p_u^{\Bbbk} f_{S,u}^{\Bbbk} S_u^{\Bbbk}(t)\,P_u(t)\,d\Gamma_{SI,u}^{\Bbbk} - \theta_{I,u}^{\Bbbk} I_u^{\Bbbk}(t) d\Gamma_{IP,u}^{\Bbbk} - \delta\, I_u^{\Bbbk}(t)\,dt, \tag{S33}$$

$$dJ_u^{\Bbbk}(t) = r_u^{\Bbbk} f_{S,u}^{\Bbbk} F_u(t)\,S_u^{\Bbbk}(t)\,d\Gamma_{SJ,u}^{\Bbbk} - \left\{\theta_{J,u}^{\Bbbk} + \delta\right\} J_u^{\Bbbk}(t)\,dt - \lambda_{J,u}^{\Bbbk} J_u^{\Bbbk}(t)dt, \tag{S34}$$

$$dP_u(t) = \sum_{\Bbbk\in\{\mathbb{n},\mathbb{i}\}} \Big(\beta_u^{\Bbbk}\,\theta_{I,u}^{\Bbbk}\,I_u^{\Bbbk}(t) d\Gamma_{IP,u}^{\Bbbk} - f_{S,u}^{\Bbbk}\big\{S_u^{\Bbbk}(t) + \xi_u I_u^{\Bbbk}(t)\big\} P_u(t)dt\Big) - \theta_{P,u}\,P_u(t)\,dt, \tag{S35}$$

$$\text{(S36)}\quad dF_u(t) = -\sum_{\Bbbk\in\{\Bbbk n,\Bbbk i\}} f^{\Bbbk}_{S,u}\, F_u(t) \left(S^{\Bbbk}_u(t) + \xi_{J,u}\, J^{\Bbbk}_u(t) + \xi_u\, I^{\Bbbk}_u(t) \right) dt + \mu\, dt + F_u(t)\, d\Gamma_{F,u},$$

$$\text{(S37)}\quad d\Gamma^{\Bbbk}_{JS,u} \sim \Gamma\left[\frac{dt}{(\sigma^{\Bbbk}_{JS,u})^2}, (\sigma^{\Bbbk}_{JS,u})^2\right], \quad d\Gamma^{\Bbbk}_{SI,u} \sim \Gamma\left[\frac{dt}{(\sigma^{\Bbbk}_{SI,u})^2}, (\sigma^{\Bbbk}_{SI,u})^2\right],$$

$$\text{(S38)}\quad d\Gamma^{\Bbbk}_{IP,u} \sim \Gamma\left[\frac{dt}{(\sigma^{\Bbbk}_{IP,u})^2}, (\sigma^{\Bbbk}_{IP,u})^2\right], \quad d\Gamma_{F,u} \sim \mathrm{N}\big[0, \sigma^2_{F,u}\, dt\big].$$

S15. Tutorials on conducting the data analysis in pomp and Pypomp. To assist other scientists and statisticians in carrying out similar investigations, we have provided tutorials that guide the reader through practical issues arising in the data analysis. We initially carried out the data analysis using the pomp R package (King et al., 2016). We later re-implemented the analysis using the Pypomp Python library (Abkemeier et al., 2026). Pypomp can compile to CPU or GPU hardware, since it is written in JAX (Bradbury et al., 2018). After checking that the two implementations are in close numerical agreement, we used Pypomp for the simulation studies in Sections S3 and S4.

A tutorial on the pomp implementation of our ecological model and data analysis is available at https://pypomp.github.io/Daphnia-tutorial/R-code/. A similar tutorial, written in Python/JAX using Pypomp, is available at https://pypomp.github.io/Daphnia-tutorial/Python-code/. Source code for these tutorials is at https://github.com/pypomp/Daphnia-tutorial (Yang et al., 2026).

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